Numerical and Statistical Methods for Finance

A. Statistical Modeling: Descriptive Statistics and Statistical Inference

1. Descriptive statistics vs statistical inference (population vs sample).
2. Sample size, random sample, realization of random variables.
3. Modeling assumptions:
   (a) Randomness assumption.
   (b) i.i.d., Markov chain, exchangeability.
   (c) Parametric model F_θ, parameter space Θ.
4. Goals of inference: estimation, testing, confidence sets.
5. Frequentist vs Bayesian approaches (prior, posterior).
6. Statistics/transformations $T : \mathbb{R}^n \to \mathbb{R}^m$.
7. Model types: continuous vs discrete.

B. Probability Distributions

B1. Continuous models

1. Gaussian/Normal distribution
   (a) Linear transformations.
   (b) Closure under convolution (independent sums).
   (c) Linear combinations of independent Gaussians.
2. Uniform distribution.
3. Probability integral transform.
4. Gamma function.
5. Gamma distribution (additivity for common β).
6. Exponential distribution (as Gamma).
7. Beta distribution + Gamma-ratio construction.

B2. Discrete models

1. Bernoulli distribution.
2. Binomial distribution (sum of Bernoulli; additivity).
3. Hypergeometric distribution.
4. Poisson distribution (additivity).
5. Mentioned links / connections
   (a) Geometric.
   (b) Negative Binomial.
   (c) Poisson $\leftrightarrow$ Negative Binomial / Geometric (Gamma–Poisson mixture).
   (d) Beta–Binomial (Beta mixing of Binomial).

C. Random Samples and Basic Statistics

1. i.i.d. sampling; joint density/pmf factorization.
2. Statistics as dimension reduction.
3. Common statistics: mean, variances, moments, min/max.
4. Order statistics distributions (min, max).
5. Gaussian sample mean: distribution, expectation, variance.

D. χ^2, t, and F Distributions; Sample Variance

1. χ^2_k distribution (Gamma representation).
2. Additivity of independent χ^2; scaling.
3. Student t_k distribution.
4. $F(m, n)$ distribution.
5. Chi-square splitting; sums of squares decomposition.
6. Distribution of corrected sample variance S^2.

E. Likelihood and Maximum Likelihood Estimation (MLE)

1. Likelihood & log-likelihood (continuous/discrete).
2. Likelihood equation; numerical issues.
3. Bernoulli model: $\widehat{\theta}_{\mathrm{ML}} = \bar{X}_n$.
4. Regularity conditions for likelihood.

F. Identifiability, Score Function, Fisher Information

1. Identifiability.
2. Score function $V_n'(\theta)$.
3. $\mathbb{E}[V_n'(\theta)] = 0$ (under regularity).
4. Fisher information identities and scaling $I_n(\theta) = nI_1(\theta)$.

G. Exponential Family (EF)

1. EF general form.
2. Bernoulli as EF example.
3. Mean identity in EF.
4. Variance identity in EF.
5. Natural parameterization.
6. MLE in natural parametrization.
7. EF, MLE dependence on $\sum A(X_i)$, sufficiency.

H. Sufficiency and Factorization

1. Definition via conditional distribution.
2. Bernoulli example: $T_n = \sum X_i$ sufficient.
3. Discrete conditioning reminder.
4. Fisher ratio criterion.
5. Fisher–Neyman factorization theorem (+ proof sketch both directions).
6. Applications: Gaussian mean (σ known), EF, trivial sufficiency, bijection invariance, Beta counterexample, Uniform $U(a-b, a+b)$.

I. Minimal Sufficiency

1. Definition (functional dependence / partitions).
2. Likelihood ratio characterization.
3. Lehmann–Scheffé theorem (statement + proof structure).
4. Examples: Bernoulli; Gaussian mean (σ known).

J. Properties of Estimators

1. Finite-sample vs asymptotic properties.
2. Unbiasedness and bias $b_n(\theta)$.
3. Unbiasedness of sample mean.
4. Bias of uncorrected variance; corrected variance unbiased.
5. MSE decomposition: MSE = Var + bias^2.
6. Relative efficiency via MSE.

K. Cramér–Rao Bound and Efficient Estimation

1. CRLB (unbiased case) + proof.
2. Biased CRLB and MSE lower bound.
3. Efficiency: definition, uniqueness, characterization.
4. Link with exponential family; score affine in statistic.
5. Newton–Raphson / Fisher scoring update.

L. Rao–Blackwell and Asymptotic Concepts

1. Rao–Blackwell theorem (statement + proof outline).
2. Variance decomposition.
3. Asymptotic notions: asymptotically correct, consistency (MS $\Rightarrow$ P), strong consistency, asymptotic efficiency, asymptotic normality.

M. Exercises and the Bayesian Approach

1. Bernoulli: MLE, unbiasedness, variance, Fisher info, efficiency, consistency, CLT.
2. Beta–Bernoulli conjugacy; Bayes estimator (squared loss).
3. Continuous model $f(x;\theta) = 2\theta x e^{-\theta x^2}$: MLE, MOM, sufficiency, Bayes with Exp prior.
4. Uniform $U(a-b, a+b)$: likelihood constraints, sufficiency, MOM, MLE via order stats.

N. MLE Properties

1. Basic properties: sufficiency, invariance, consistency, Cramér theorem (asymptotic normality).
2. Invariance principle examples: survival/binomial reparametrization; misclassification two-group model.
3. Weak vs strong consistency (definitions).
4. Remark: MLEs typically strongly consistent + examples.
5. Strong consistency sketch via log-likelihood (finite Θ union bound argument).

O. o_p, O_p, Taylor in Probability, Delta Method, Cramér Theorem

1. Definitions of $o_p(\tau_n)$, $O_p(\tau_n)$ (incl. $\tau_n = 1$).
2. Algebra rules for o_p, O_p.
3. Convergence in distribution $\Rightarrow O_p$ bound.
4. Taylor expansion in probability.
5. Delta method (+ application to MLE transforms).
6. Cramér theorem: statement, Taylor of score, CLT/LLN/remainder, consequences.

P. Hypothesis Testing: Basics

1. Testing setting and goal.
2. Hypotheses: simple vs composite (one/two-sided).
3. Decision rule and rejection region.
4. Type I/II errors, power.
5. Power function; size/level.
6. UMP in a class; design principles.
7. Binomial example (α–β trade-off).
8. Trade-off proof via inclusion of rejection regions.

Q. Neyman–Pearson (NP) Lemma, Consequences, and One-Parameter Structure

1. Neyman–Pearson lemma (simple vs. simple) + proof + uniqueness (up to the boundary).
2. NP examples: Normal mean (variance known) and Bernoulli (discrete calibration / randomization).
3. One-parameter exponential family: LR monotonicity and threshold rule in $T = \sum_{i=1}^{n} A(X_i)$.
4. NP with sufficient statistic (factorization consequence): LR depends on T only.
5. Karlin–Rubin (MLR) corollary: one-sided composite UMP tests via threshold in T.

R. Likelihood Ratio Test (LRT): Finite-Sample Forms and Examples

1. Definition of the LRT: $\Lambda(x) = \dfrac{\sup_{\theta\in\Theta_0} L(\theta; x)}{\sup_{\theta\in\Theta} L(\theta; x)}$, reject for small Λ.
2. Classical finite-sample LRTs in Gaussian models (exact reductions):
   (a) One-sample normal (unknown mean and variance): LRT $\Rightarrow$ two-sided t-test.
   (b) Two-sample normal (equal variances): LRT for equality of means $\Rightarrow$ pooled two-sample t-test.
   (c) Two-sample normal: LRT for equality of variances $\Rightarrow$ F-test.
3. Why Gaussianity matters for exact finite-sample $t/F/\chi^2$ pivots.

S. Asymptotic LRT: Wilks Theorem

1. LRT setup: likelihood ratio statistic; reject for small values; calibration via quantiles.
2. Wilks' theorem (regular models, general/multidimensional): chi-square limit under H_0 with df = #constraints (proof included here).
3. Poisson LRT example.
4. Non-regular cautionary example: Wilks may fail when support depends on the parameter.
5. Multidimensional Wilks: df via difference in free parameters; examples (linear model; multinomial/coin-flip constraints).
6. Wald test: asymptotic normality-based test.
7. Consistency and power (asymptotic size and power $\to 1$).
8. Example: Wald test for Bernoulli (one-sided, plug-in standard error).

A. Statistical Modeling: Descriptive Statistics and Statistical Inference

1. **Descriptive statistics vs. statistical inference (population vs. sample).** Given observed data $(x_1, \ldots, x_n)$, *descriptive statistics* aim to summarize and represent these numbers (e.g. through averages, dispersion measures, and graphs) without extending conclusions beyond the dataset at hand. *Statistical inference* aims to learn features of the underlying *population* from which the data were generated, using a *sample* and quantifying the uncertainty induced by sampling variability.
2. **Sample size, random sample, realization of random variables.** The integer $n \geq 1$ is the *sample size*. The vector $(x_1, \ldots, x_n) \in \mathbb{R}^n$ is an observed *sample*. In the modeling step, one assumes that the observed sample is a *realization* of a random vector $(X_1, \ldots, X_n)$ defined on a probability space. A *random sample* typically refers to the collection $(X_1, \ldots, X_n)$ under a sampling assumption (often i.i.d.).
3. **Modeling assumptions.**
   (a) **Randomness assumption ($X_1, \ldots, X_n$ underlying $x_1, \ldots, x_n$).** One postulates that the observed data $(x_1, \ldots, x_n)$ arise as a realization of random variables $(X_1, \ldots, X_n)$. This assumption turns the data into outcomes of a probabilistic experiment and makes it possible to define probabilities of events and sampling distributions of statistics.
   (b) **Symmetry/dependence structures: i.i.d., Markov chain, exchangeability.** A weak form of structure is to assume that the variables are *symmetric* or have a specified dependence pattern. Common cases include: (i) *i.i.d.* sampling, where $X_1, \ldots, X_n$ are independent and identically distributed; (ii) *Markov* dependence, where the conditional distribution of X_{t+1} given the past depends only on X_t; (iii) *exchangeability*, where the joint distribution is invariant under permutations of the indices, capturing a notion of symmetry without full independence.
   (c) **Parametric statistical model F_θ, parameter space Θ (finite-dimensional).** A strong assumption is to specify a parametric family of distributions $\{F_\theta : \theta \in \Theta\}$ for the data-generating mechanism. A standard setup is

$$X_1, \ldots, X_n \overset{iid}{\sim} F_\theta, \qquad \theta \in \Theta,$$

   where Θ is a *finite-dimensional* parameter space. The parameter θ encodes the unknown characteristics of the population (e.g. mean, variance, intensity, success probability).
4. **Goals of inference: estimation, testing, confidence sets.** The objective is to learn the unknown parameter (or functionals of it). Core tasks are: (i) *estimation*, producing a point estimate $\widehat{\theta}$ (or $\widehat{\tau} = \tau(\widehat{\theta})$); (ii) *hypothesis testing*, deciding whether the data are compatible with a null restriction (e.g. $\theta \in \Theta_0$); (iii) *confidence sets*, constructing random sets $C(X_1, \ldots, X_n)$ that contain the true parameter with a prescribed probability (coverage) under repeated sampling.
5. **Frequentist vs Bayesian approaches (prior G, posterior $\theta \mid x_1, \ldots, x_n$).** In the *frequentist* approach, θ is a fixed (unknown) constant and randomness comes only from the sample; inference is evaluated by repeated-sampling properties (e.g. bias, variance, coverage, error probabilities). In the *Bayesian* approach, θ is modeled as a random variable with *prior* distribution G; given data,

$$X_1, \ldots, X_n \mid \theta \overset{iid}{\sim} F_\theta, \qquad \theta \sim G,$$

   and learning is summarized by the *posterior* distribution $\theta \mid x_1, \ldots, x_n$ obtained via Bayes' rule.
6. **Statistics/transformations $T : \mathbb{R}^n \to \mathbb{R}^m$ ($m \leq n$).** A *statistic* is any measurable function of the sample,

$$T = T(X_1, \ldots, X_n), \qquad T : \mathbb{R}^n \to \mathbb{R}^m, \quad m \leq n,$$

   used to reduce dimension and extract information relevant for inference (e.g. $\bar{X}$, S^2, order statistics). After observation, the statistic takes the value $T(x_1, \ldots, x_n)$.
7. **Model types: continuous vs discrete.** The model F_θ may be *continuous* (with density $f(\cdot; \theta)$ on $\mathbb{R}$ or $\mathbb{R}^d$) or *discrete* (with probability mass function $p(\cdot; \theta)$ on a countable support, e.g. $\mathbb{N}_0$). This distinction affects likelihood expressions, integration vs summation, and the form of sampling distributions.

B. Probability Distributions

B1. Continuous models

1. **Gaussian/Normal distribution.** A random variable Y has a *normal* distribution with mean $\mu \in \mathbb{R}$ and variance $\sigma^2 > 0$, written $Y \sim \mathcal{N}(\mu, \sigma^2)$, if it has density

$$f_Y(t; \mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left\{ -\frac{(t-\mu)^2}{2\sigma^2} \right\} \mathbb{1}_{\mathbb{R}}(t).$$

The *standard normal* is $\mathcal{N}(0,1)$; its density and CDF are

$$\phi(t) = \frac{1}{\sqrt{2\pi}} e^{-t^2/2}, \qquad \Phi(t) = \int_{-\infty}^{t} \phi(x)\, dx.$$

(a) **Linear transformations.** If $Y \sim \mathcal{N}(\mu, \sigma^2)$ and $a, b \in \mathbb{R}$, then

$$a + bY \sim \mathcal{N}(a + b\mu,\ b^2\sigma^2).$$

(b) **Closure under convolution (independent sums).** If $Y_1 \sim \mathcal{N}(\mu_1, \sigma_1^2)$, $Y_2 \sim \mathcal{N}(\mu_2, \sigma_2^2)$ and $Y_1 \perp\!\!\!\perp Y_2$, then

$$Y_1 + Y_2 \sim \mathcal{N}(\mu_1 + \mu_2,\ \sigma_1^2 + \sigma_2^2).$$

(c) **Linear combinations of independent Gaussians.** If $Y_1, \ldots, Y_n$ are independent with $Y_i \sim \mathcal{N}(\mu_i, \sigma_i^2)$ and $a_i, b \in \mathbb{R}$, then

$$\sum_{i=1}^{n} a_i Y_i + b \sim \mathcal{N}\left(\sum_{i=1}^{n} a_i \mu_i + b,\ \sum_{i=1}^{n} a_i^2 \sigma_i^2 \right).$$

2. **Uniform distribution.** A random variable Y has a *uniform* distribution on (a, b), written $Y \sim U(a, b)$ with $a < b$, if

$$f_Y(t; a, b) = \frac{1}{b-a}\, \mathbb{1}_{(a,b)}(t) = \begin{cases} \frac{1}{b-a}, & t \in (a, b), \\ 0, & \text{otherwise.} \end{cases}$$

3. **Probability integral transform.** Let Z be a continuous random variable with CDF F. Then

$$W = F(Z) \sim U(0, 1).$$

In particular, for $t \in [0, 1]$,

$$\mathbb{P}(W \le t) = \mathbb{P}(F(Z) \le t) = \mathbb{P}(Z \le F^{-1}(t)) = F(F^{-1}(t)) = t.$$

4. **Gamma function.** For $x > 0$, the Gamma function is defined by

$$\Gamma(x) = \int_0^{+\infty} t^{x-1} e^{-t}\, dt.$$

Key identities:

$$\Gamma(x+1) = x\,\Gamma(x), \qquad \Gamma(n) = (n-1)!\ (n \in \mathbb{N}), \qquad \Gamma\left(\tfrac{1}{2}\right) = \sqrt{\pi}.$$

5. **Gamma distribution.** A random variable Y has a *Gamma* distribution with parameters $\alpha > 0$ and $\beta > 0$, written $Y \sim \text{Gamma}(\alpha, \beta)$, if

$$f_Y(t; \alpha, \beta) = \frac{\beta^\alpha}{\Gamma(\alpha)}\, t^{\alpha-1} e^{-\beta t}\, \mathbb{1}_{\mathbb{R}^+}(t).$$

(*Additivity for common β.*) If $Y_1 \sim \text{Gamma}(\alpha_1, \beta)$ and $Y_2 \sim \text{Gamma}(\alpha_2, \beta)$ with $Y_1 \perp\!\!\!\perp Y_2$, then

$$Y_1 + Y_2 \sim \text{Gamma}(\alpha_1 + \alpha_2, \beta).$$

6. **Exponential distribution (as a Gamma).** The *(negative) exponential* distribution with rate $\lambda > 0$ is

$$Y \sim \text{Exp}(\lambda) \quad \Longleftrightarrow \quad Y \sim \text{Gamma}(1, \lambda),$$

with density

$$f_Y(t; \lambda) = \lambda e^{-\lambda t} \, \mathbb{1}_{\mathbb{R}^+}(t).$$

If $Y_1, \ldots, Y_n$ are independent and $Y_i \sim \text{Exp}(\lambda)$, then

$$\sum_{i=1}^{n} Y_i \sim \text{Gamma}(n, \lambda).$$

7. **Beta distribution and Gamma-ratio construction.** A random variable Y has a *Beta* distribution with parameters $a > 0$ and $b > 0$, written $Y \sim \text{Beta}(a, b)$, if

$$f_Y(t; a, b) = \frac{\Gamma(a+b)}{\Gamma(a)\Gamma(b)} \, t^{a-1}(1-t)^{b-1} \, \mathbb{1}_{(0,1)}(t).$$

$$\mathbb{E}[X] = \frac{a}{a+b} \qquad \text{and} \qquad \text{Var}(X) = \frac{ab}{(a+b)^2 \, (a+b+1)}.$$

(*Construction via independent Gammas.*) If $Y_1 \sim \text{Gamma}(a, \lambda)$, $Y_2 \sim \text{Gamma}(b, \lambda)$ are independent (same λ), then

$$\frac{Y_1}{Y_1 + Y_2} \sim \text{Beta}(a, b).$$

B2. Discrete models

1. **Bernoulli distribution.** A random variable Y has a *Bernoulli* distribution with parameter $\theta \in (0, 1)$, written $Y \sim \text{Ber}(\theta)$, if

$$\mathbb{P}(Y = x) = p_Y(x; \theta) = \theta^x (1-\theta)^{1-x} \, \mathbb{1}_{\{0,1\}}(x), \qquad x \in \{0, 1\}.$$

2. **Binomial distribution.** A random variable Y has a *Binomial* distribution with parameters $n \in \mathbb{N}$ and $\theta \in (0, 1)$, written $Y \sim \text{Bin}(n, \theta)$, if

$$\mathbb{P}(Y = x) = p_Y(x; n, \theta) = \binom{n}{x} \theta^x (1-\theta)^{n-x} \, \mathbb{1}_{\{0,\ldots,n\}}(x).$$

(*Sum of i.i.d. Bernoulli.*) If $Y_1, \ldots, Y_n \overset{iid}{\sim} \text{Ber}(\theta)$, then

$$\sum_{i=1}^{n} Y_i \sim \text{Bin}(n, \theta).$$

(*Additivity in n for common θ.*) If $Y_1 \sim \text{Bin}(n_1, \theta)$, $Y_2 \sim \text{Bin}(n_2, \theta)$ and $Y_1 \perp\!\!\!\perp Y_2$, then

$$Y_1 + Y_2 \sim \text{Bin}(n_1 + n_2, \theta).$$

3. **Hypergeometric distribution.** A random variable Y has a *Hypergeometric* distribution with parameters $(\mu, N, M) \in \mathbb{N}^3$, written $Y \sim \text{Hyp}(\mu, N, M)$, if

$$\mathbb{P}(Y = x) = \frac{\binom{M}{x}\binom{N-M}{\mu-x}}{\binom{N}{\mu}} \, \mathbb{1}_{\{\max(0, \mu-N+M), \ldots, \min(\mu, M)\}}(x).$$

4. **Poisson distribution.** A random variable Y has a *Poisson* distribution with parameter $\lambda > 0$, written $Y \sim \text{Po}(\lambda)$, if

$$\mathbb{P}(Y = x) = p_Y(x; \lambda) = \frac{e^{-\lambda} \lambda^x}{x!} \, \mathbb{1}_{\mathbb{N}_0}(x), \qquad x \in \mathbb{N}_0.$$

(*Additivity.*) If $Y_1 \sim \text{Po}(\lambda_1)$, $Y_2 \sim \text{Po}(\lambda_2)$ and $Y_1 \perp\!\!\!\perp Y_2$, then

$$Y_1 + Y_2 \sim \text{Po}(\lambda_1 + \lambda_2).$$

5. **Mentioned links (CHATGPT).** Connections mentioned in the notes include: Poisson–Binomial and Poisson–Negative Binomial / Geometric. (Only referenced here; details are developed elsewhere in the notes.

   (a) **Geometric (Bernoulli trials until the first success).** Let $(B_i)_{i\geq 1}$ be i.i.d. Bernoulli(p) trials. Define
   $$T := \min\{k \geq 1 : B_k = 1\},$$
   the number of trials up to (and including) the first success. Then $T \in \{1, 2, \dots\}$ and its pmf is
   $$\mathbb{P}(T = k) = (1-p)^{k-1}p, \qquad k = 1, 2, \dots$$
   (Equivalently, if F denotes the number of failures before the first success, then $F = T - 1 \in \{0, 1, 2, \dots\}$ and $\mathbb{P}(F = n) = (1-p)^n p$.)

   (b) **Negative Binomial (failures before the r-th success).** Let $(B_i)_{i\geq 1}$ be i.i.d. Bernoulli(p) and let N be the number of failures observed before the r-th success (with $r > 0$). Then $N \in \{0, 1, 2, \dots\}$ and
   $$\mathbb{P}(N = n) = \binom{n+r-1}{n}(1-p)^n p^r, \qquad n = 0, 1, 2, \dots$$
   with the generalized binomial coefficient
   $$\binom{n+r-1}{n} = \frac{\Gamma(n+r)}{\Gamma(r)\,\Gamma(n+1)} \qquad (r > 0).$$
   In particular, the geometric law is recovered as the special case $r = 1$ (failures before the first success).

   (c) **Poisson $\leftrightarrow$ Negative Binomial / Geometric (Gamma–Poisson mixture).** If $\Lambda \sim \text{Gamma}(r, \lambda)$ and, conditional on Λ, $N \mid \Lambda \sim \text{Po}(\Lambda)$, then
   $$N \sim \text{NegBin}(r,\ p), \qquad p = \frac{\lambda}{\lambda+1},$$
   in a standard parameterization. In particular, the *geometric* distribution is obtained as the special case $r = 1$.

   (d) **Beta–Binomial (Beta mixing of Binomial).** If $\Theta \sim \text{Beta}(a, b)$ and, conditional on Θ, $Y \mid \Theta \sim \text{Bin}(n, \Theta)$, then
   $$Y \sim \text{BetaBin}(n, a, b),$$
   with pmf (for $y = 0, 1, \dots, n$)
   $$\mathbb{P}(Y = y) = \binom{n}{y}\frac{B(a+y,\ b+n-y)}{B(a,b)} = \binom{n}{y}\frac{\Gamma(a+y)\Gamma(b+n-y)\Gamma(a+b)}{\Gamma(a)\Gamma(b)\Gamma(a+b+n)}.$$

C. Random Samples and Basic Statistics

1. **i.i.d. sampling from F_θ; joint density/mass factorization.** Let $X_1, \dots, X_n$ be a *random sample* from a parametric model $\{F_\theta : \theta \in \Theta\}$, i.e.
   $$X_1, \dots, X_n \overset{iid}{\sim} F_\theta, \qquad \theta \in \Theta.$$
   Denote $\underline{X} = (X_1, \dots, X_n)$ and $\underline{x} = (x_1, \dots, x_n)$.
   - *Continuous case:* if F_θ admits a density $f_X(\cdot;\theta)$, then the joint density factorizes as
   $$f_{\underline{X}}(\underline{x};\theta) = \prod_{i=1}^{n} f_X(x_i;\theta).$$

- *Discrete case:* if F_θ admits a pmf $p_X(\cdot;\theta)$, then the joint pmf factorizes as

$$p_{\underline{X}}(\underline{x};\theta) = \prod_{i=1}^{n} p_X(x_i;\theta).$$

This factorization is the key structural consequence of independence and identical distribution.

2. **Statistics $T(X_1,\dots,X_n)$ as dimension reduction.** A *statistic* is any (measurable) function of the sample,

$$T = T(\underline{X}) = T(X_1,\dots,X_n), \qquad T:\mathbb{R}^n \to \mathbb{R}^m, \quad m \le n.$$

It reduces the data from dimension n to m while retaining information relevant to the inferential goal (e.g. estimating θ, testing hypotheses, or building confidence sets).

3. **Common statistics.** For a univariate sample $(X_1,\dots,X_n)$:
   (a) **Sample mean.**

$$\bar{X}_n = \frac{1}{n}\sum_{i=1}^{n} X_i.$$

   (b) **Sample variance (uncorrected).**

$$\tilde{S}_n^2 = \frac{1}{n}\sum_{i=1}^{n}(X_i - \bar{X}_n)^2.$$

   (c) **Corrected sample variance.**

$$S_n^2 = \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \bar{X}_n)^2.$$

   (d) **Sample moments.** For an integer $r \ge 1$,

$$M_{r,n} = \frac{1}{n}\sum_{i=1}^{n} X_i^r.$$

   (e) **Order statistics: minimum and maximum.**

$$X_{(1)} = \min\{X_1,\dots,X_n\}, \qquad X_{(n)} = \max\{X_1,\dots,X_n\}.$$

4. **Distributions of order statistics (i.i.d. case).** Assume $X_1,\dots,X_n$ are i.i.d. with CDF F. If F has density f, then:
   (a) **Maximum $X_{(n)}$.**

$$\mathbb{P}(X_{(n)} \le t) = \mathbb{P}(X_1 \le t,\dots,X_n \le t) = \prod_{i=1}^{n}\mathbb{P}(X_i \le t) = (F(t))^n,$$

and (when f exists)

$$f_{X_{(n)}}(t) = \frac{d}{dt}(F(t))^n = n(F(t))^{n-1}f(t).$$

   (b) **Minimum $X_{(1)}$.**

$$\mathbb{P}(X_{(1)} \ge t) = \mathbb{P}(X_1 \ge t,\dots,X_n \ge t) = \prod_{i=1}^{n}\mathbb{P}(X_i \ge t) = (1-F(t))^n,$$

hence

$$\mathbb{P}(X_{(1)} \le t) = 1 - (1-F(t))^n,$$

and (when f exists)

$$f_{X_{(1)}}(t) = \frac{d}{dt}\Big(1-(1-F(t))^n\Big) = n(1-F(t))^{n-1}f(t).$$

5. **Gaussian sample mean: distribution, expectation, variance.** If $X_1, \ldots, X_n \overset{iid}{\sim} \mathcal{N}(\mu, \sigma^2)$, then by closure of the Gaussian law under linear combinations,

$$\bar{X}_n = \frac{1}{n}\sum_{i=1}^{n} X_i \sim \mathcal{N}\left(\mu, \frac{\sigma^2}{n}\right).$$

Moreover,

$$\mathbb{E}[\bar{X}_n] = \mathbb{E}\left[\frac{1}{n}\sum_{i=1}^{n} X_i\right] = \mu, \qquad \mathrm{Var}(\bar{X}_n) = \mathrm{Var}\left(\frac{1}{n}\sum_{i=1}^{n} X_i\right) = \frac{\sigma^2}{n}.$$

D. χ^2, t, and F Distributions; Sample Variance

1. χ^2_k **distribution (sum of squares; Gamma representation).** Let $Z_1, \ldots, Z_k$ be i.i.d. standard Gaussians:

$$Z_i \sim N(0,1), \qquad Z_i \perp\!\!\!\perp Z_j \ (i \neq j).$$

Define

$$Z = \sum_{i=1}^{k} Z_i^2.$$

Then Z has a chi-square distribution with k degrees of freedom:

$$Z \sim \chi^2_k.$$

Density:

$$\left[f_Z(z;k) = \frac{\left(\frac{1}{2}\right)^{k/2}}{\Gamma(k/2)}\, z^{k/2-1} \exp\left\{-\frac{1}{2}z\right\} \mathbb{1}_{\mathbb{R}^+}(z) \right].$$

Hence the Gamma representation:

$$Z \sim \mathrm{Gamma}\left(\frac{k}{2}, \frac{1}{2}\right).$$

2. **Additivity of independent χ^2 variables; scaling remarks.** If

$$Z_1 \sim \chi^2_k, \qquad Z_2 \sim \chi^2_\delta, \qquad Z_1 \perp\!\!\!\perp Z_2,$$

then

$$Z_1 + Z_2 \sim \chi^2_{k+\delta}.$$

Scaling (via Gamma form): if $Z \sim \chi^2_k$ and $c > 0$, then

$$cZ \sim \mathrm{Gamma}\left(\frac{k}{2}, \frac{1}{2c}\right).$$

3. **Student t_k distribution.** Let

$$Z \sim N(0,1), \qquad V \sim \chi^2_k, \qquad Z \perp\!\!\!\perp V.$$

Define

$$T = \frac{Z}{\sqrt{V/k}}.$$

Then

$$T \sim t_k,$$

with density

$$\left[f_T(t;k) = \frac{\Gamma\left(\frac{k+1}{2}\right)}{\sqrt{k\pi}\,\Gamma\left(\frac{k}{2}\right)} \left(1 + \frac{t^2}{k}\right)^{-(k+1)/2} \mathbb{1}_{\mathbb{R}}(t) \right].$$

4. $F(m,n)$ **distribution.** Let

$$X \sim \chi^2_m, \qquad Y \sim \chi^2_n, \qquad X \perp\!\!\!\perp Y, \qquad m, n \in \mathbb{N}.$$

Define

$$F = \frac{(X/m)}{(Y/n)}.$$

Then

$$F \sim F(m,n).$$

5. **Chi-square splitting and sums of squares decomposition.**
Theorem (chi-square splitting). Let Q, Q_1, Q_2 be random variables such that

$$Q = Q_1 + Q_2.$$

Assume

$$Q \sim \chi^2_g, \qquad Q_1 \sim \chi^2_{g_1}, \qquad g_2 := g - g_1 \geq 0, \qquad \text{and} \qquad Q_1 \perp Q_2.$$

Then

$$Q_2 \sim \chi^2_{g_2}.$$

Let $X_1, \ldots, X_n \overset{\text{i.i.d.}}{\sim} N(\mu, \sigma^2)$ and define

$$\bar{X} = \frac{1}{n}\sum_{i=1}^n X_i, \qquad S^2 = \frac{1}{n-1}\sum_{i=1}^n (X_i - \bar{X})^2.$$

Set

$$Z_i := \frac{X_i - \mu}{\sigma}, \qquad i = 1, \ldots, n,$$

so that $Z_i \sim N(0,1)$ i.i.d. and hence

$$Q := \sum_{i=1}^n Z_i^2 = \sum_{i=1}^n \left(\frac{X_i - \mu}{\sigma}\right)^2 \sim \chi^2_n.$$

Using $X_i - \mu = (X_i - \bar{X}) + (\bar{X} - \mu)$ and $\sum_{i=1}^n (X_i - \bar{X}) = 0$, we obtain

$$\sum_{i=1}^n \left(\frac{X_i - \mu}{\sigma}\right)^2 = \sum_{i=1}^n \left(\frac{X_i - \bar{X}}{\sigma}\right)^2 + n\left(\frac{\bar{X} - \mu}{\sigma}\right)^2.$$

Moreover, since $\bar{X} \sim N(\mu, \sigma^2/n)$,

$$Q_1 := n\left(\frac{\bar{X} - \mu}{\sigma}\right)^2 = \left(\frac{\bar{X} - \mu}{\sigma/\sqrt{n}}\right)^2 \sim \chi^2_1.$$

Define

$$Q_2 := \sum_{i=1}^n \left(\frac{X_i - \bar{X}}{\sigma}\right)^2.$$

In the Gaussian model one has $Q_1 \perp Q_2$, so

$$Q_2 \sim \chi^2_{n-1}, \qquad \text{i.e.} \qquad \sum_{i=1}^n \left(\frac{X_i - \bar{X}}{\sigma}\right)^2 \sim \chi^2_{n-1}.$$

6. **Distribution of the corrected sample variance.** Since $\sum_{i=1}^n (X_i - \bar{X})^2 = (n-1)S^2$, we have

$$\sum_{i=1}^n \left(\frac{X_i - \bar{X}}{\sigma}\right)^2 = \frac{(n-1)S^2}{\sigma^2}.$$

Therefore,

$$\boxed{\frac{n-1}{\sigma^2} S^2 \sim \chi^2_{n-1}}.$$

Equivalently,

$$S^2 \sim \frac{\sigma^2}{n-1}\chi^2_{n-1}.$$

E. Likelihood and Maximum Likelihood Estimation (MLE)

1. **Likelihood and log-likelihood for i.i.d. samples (continuous and discrete).**
   Let $\underline{X} = (X_1, \ldots, X_n)$ be a random sample from a parametric model F_θ, $\theta \in \Theta$.
   - **Continuous case.** If X_i have density $f_X(\cdot;\theta)$, then the joint density factorizes as

$$f_{\underline{X}}(\underline{x};\theta) = \prod_{i=1}^{n} f_X(x_i;\theta).$$

     The **likelihood function** is the same expression viewed as a function of θ for fixed data $\underline{x}$:

$$\mathcal{L}(\theta;\underline{x}) = \prod_{i=1}^{n} f_X(x_i;\theta).$$

     The **log-likelihood** is

$$V_n(\theta) := \log \mathcal{L}(\theta;\underline{x}) = \sum_{i=1}^{n} \log f_X(x_i;\theta).$$

   - **Discrete case.** If X_i have pmf $p_X(\cdot;\theta)$, then

$$P_{\underline{X}}(\underline{x};\theta) = \prod_{i=1}^{n} p_X(x_i;\theta), \qquad \mathcal{L}(\theta;\underline{x}) = \prod_{i=1}^{n} p_X(x_i;\theta),$$

     and

$$V_n(\theta) = \log \mathcal{L}(\theta;\underline{x}) = \sum_{i=1}^{n} \log p_X(x_i;\theta).$$

   The **MLE** is any maximizer of $\mathcal{L}(\theta;\underline{x})$ (equivalently of $V_n(\theta)$):

$$\hat{\theta}_{\mathrm{ML}} \in \arg\max_{\theta\in\Theta} \mathcal{L}(\theta;\underline{x}) \qquad \Longleftrightarrow \qquad \hat{\theta}_{\mathrm{ML}} \in \arg\max_{\theta\in\Theta} V_n(\theta).$$

2. **Likelihood equation and numerical issues (no closed-form in general).**
   In regular problems, an interior maximizer $\hat{\theta}$ typically satisfies the **likelihood equation**

$$\frac{d}{d\theta} V_n(\theta) = \frac{d}{d\theta} \log \mathcal{L}(\theta;\underline{x}) = 0,$$

   together with the second-order condition

$$\frac{d^2}{d\theta^2} V_n(\hat{\theta}) < 0.$$

   In general, the equation $\frac{d}{d\theta} \log \mathcal{L}(\theta;\underline{x}) = 0$ may not admit an explicit closed-form solution, hence numerical methods (e.g. Newton–Raphson) are used.
3. **Bernoulli model: explicit MLE derivation $\widehat{\theta}_{\mathbf{ML}} = \bar{X}_n$.**
   Let $X_1, \ldots, X_n \overset{iid}{\sim} \mathrm{Ber}(\theta)$ with $\theta \in (0,1)$ and observed sample $\underline{x} = (x_1, \ldots, x_n) \in \{0,1\}^n$. The likelihood is

$$\mathcal{L}(\theta;\underline{x}) = \prod_{i=1}^{n} \theta^{x_i}(1-\theta)^{1-x_i} = \theta^{\sum_{i=1}^n x_i}(1-\theta)^{n-\sum_{i=1}^n x_i}.$$

   The log-likelihood is

$$V_n(\theta) = \log \mathcal{L}(\theta;\underline{x}) = \Big(\sum_{i=1}^{n} x_i\Big) \log\theta + \Big(n - \sum_{i=1}^{n} x_i\Big) \log(1-\theta).$$

   Differentiate and set equal to 0:

$$V_n'(\theta) = \frac{1}{\theta}\sum_{i=1}^{n} x_i - \frac{1}{1-\theta}\Big(n - \sum_{i=1}^{n} x_i\Big) = 0,$$

hence

$$\widehat{\theta}_{\mathrm{ML}} = \frac{1}{n}\sum_{i=1}^{n} x_i = \bar{X}_n.$$

Moreover,

$$V_n''(\theta) = -\frac{\sum_{i=1}^n x_i}{\theta^2} - \frac{n - \sum_{i=1}^n x_i}{(1-\theta)^2} < 0,$$

so the critical point is a maximizer.

4. **Regularity conditions for likelihood.**
Let $(X_1, \ldots, X_n)$ be a random sample from a model with density (or mass) $f_X(\cdot;\theta)$, $\theta \in \Theta$. A set of **regularity conditions** (as in the notes) is:
   (a) Θ is an open subset of $\mathbb{R}$.
   (b) For any $\theta \in \Theta$, derivatives with respect to θ exist up to third order (for the likelihood / density as used below).
   (c) For any $\theta_0 \in \Theta$, there exist integrable functions $g(\cdot), h(\cdot), H(\cdot)$ (integrable around θ_0) such that, for all $x \in \mathbb{R}$,

$$\left|\frac{d}{d\theta} f_X(x;\theta)\right| \le g(x), \qquad \left|\frac{d^2}{d\theta^2} f_X(x;\theta)\right| \le h(x), \qquad \left|\frac{d^3}{d\theta^3} \log f_X(x;\theta)\right| \le H(x).$$

   (d) The Fisher information is finite and positive:

$$0 < \mathbb{E}\left[\left(\frac{d}{d\theta}\log \mathcal{L}(\theta;\underline{X})\right)^2\right] < +\infty, \qquad \forall \theta \in \Theta.$$

Remark. Even under regularity, the likelihood equation

$$\frac{d}{d\theta}\log \mathcal{L}(\theta;\underline{x}) = 0$$

may not be solvable analytically.

F. Identifiability, Score Function, Fisher Information

1. **Identifiability of a parametric model.**
Let $X \sim f_X(x;\theta)$ with $\theta \in \Theta$. The model is *identifiable* if distinct parameters induce distinct distributions, i.e. for any $\theta_1 \neq \theta_2$ there exists an event E such that

$$\mathbb{P}_{\theta_1}(X \in E) \neq \mathbb{P}_{\theta_2}(X \in E).$$

Equivalently, $f_X(\cdot;\theta_1) = f_X(\cdot;\theta_2)$ (a.e.) implies $\theta_1 = \theta_2$.

2. **Score function: derivative of log-likelihood $V_n'(\theta)$.**
For a random sample $\underline{X} = (X_1, \ldots, X_n)$ with realization $\underline{x} = (x_1, \ldots, x_n)$, the log-likelihood is

$$V_n(\theta) = \log \mathcal{L}(\theta;\underline{x}) = \sum_{i=1}^{n} \log f_{X_i}(x_i;\theta).$$

The *score function* is the first derivative of the log-likelihood:

$$V_n'(\theta) = \frac{d}{d\theta}\log \mathcal{L}(\theta;\underline{x}) = \frac{\mathcal{L}'(\theta;\underline{x})}{\mathcal{L}(\theta;\underline{x})}.$$

3. **Expectation of score under regularity: $\mathbb{E}[V_n'(\theta)] = 0$.**
Under the regularity conditions that justify exchanging derivative and integral,

$$\mathbb{E}[V_n'(\theta)] = \int_{\mathbb{R}^n} V_n'(\theta)\, f_{\underline{X}}(\underline{x};\theta)\, d\underline{x} = \int_{\mathbb{R}^n} \frac{f_{\underline{X}}'(\underline{x};\theta)}{f_{\underline{X}}(\underline{x};\theta)}\, f_{\underline{X}}(\underline{x};\theta)\, d\underline{x} = \int_{\mathbb{R}^n} \frac{d}{d\theta} f_{\underline{X}}(\underline{x};\theta)\, d\underline{x}$$

$$= \frac{d}{d\theta} \int_{\mathbb{R}^n} f_{\underline{X}}(\underline{x};\theta)\, d\underline{x} = \frac{d}{d\theta} 1 = 0.$$

Hence $\boxed{\mathbb{E}[V_n'(\theta)] = 0}$.

4. **Fisher information:**

(a) $I_n(\theta) = \mathrm{Var}(V_n'(\theta)) = \mathbb{E}[(V_n'(\theta))^2] = -\mathbb{E}[V_n''(\theta)]$**.**
The second derivative of the log-likelihood is

$$V_n''(\theta) = \frac{d^2}{d\theta^2} \log \mathcal{L}(\theta;\underline{x}) = \frac{f_{\underline{X}}''(\underline{x};\theta)}{f_{\underline{X}}(\underline{x};\theta)} - \left(\frac{f_{\underline{X}}'(\underline{x};\theta)}{f_{\underline{X}}(\underline{x};\theta)}\right)^2.$$

Taking expectations and using regularity:

$$\begin{aligned}
\mathbb{E}[V_n''(\theta)] &= \int_{\mathbb{R}^n} \left[\frac{f_{\underline{X}}''(\underline{x};\theta)}{f_{\underline{X}}(\underline{x};\theta)} - \left(\frac{f_{\underline{X}}'(\underline{x};\theta)}{f_{\underline{X}}(\underline{x};\theta)}\right)^2\right] f_{\underline{X}}(\underline{x};\theta)\, d\underline{x} \\
&= \int_{\mathbb{R}^n} f_{\underline{X}}''(\underline{x};\theta)\, d\underline{x} - \int_{\mathbb{R}^n} \frac{\left(f_{\underline{X}}'(\underline{x};\theta)\right)^2}{f_{\underline{X}}(\underline{x};\theta)}\, d\underline{x} \\
&= \int_{\mathbb{R}^n} f_{\underline{X}}''(\underline{x};\theta)\, d\underline{x} - \mathbb{E}[(V_n'(\theta))^2] \\
&= \frac{d^2}{d\theta^2} \int_{\mathbb{R}^n} f_{\underline{X}}(\underline{x};\theta)\, d\underline{x} - \mathbb{E}[(V_n'(\theta))^2] \\
&= 0 - \mathbb{E}[(V_n'(\theta))^2].
\end{aligned}$$

Therefore

$$\mathbb{E}[(V_n'(\theta))^2] = -\mathbb{E}[V_n''(\theta)].$$

Since $\mathbb{E}[V_n'(\theta)] = 0$, we also have

$$\mathrm{Var}(V_n'(\theta)) = \mathbb{E}[(V_n'(\theta))^2].$$

Define the *Fisher information* of the sample by

$$\boxed{I_n(\theta) := \mathrm{Var}(V_n'(\theta)) = \mathbb{E}[(V_n'(\theta))^2] = -\mathbb{E}[V_n''(\theta)].}$$

(b) **Scaling** $I_n(\theta) = nI_1(\theta)$**.**
For i.i.d. data,

$$V_n(\theta) = \sum_{i=1}^n \log f(X_i;\theta) \quad \Rightarrow \quad V_n'(\theta) = \sum_{i=1}^n \frac{d}{d\theta} \log f(X_i;\theta).$$

Let $U_i(\theta) = \frac{d}{d\theta} \log f(X_i;\theta)$. Then $U_1, \ldots, U_n$ are i.i.d. with $\mathbb{E}[U_i(\theta)] = 0$. Hence

$$I_n(\theta) = \mathrm{Var}\left(\sum_{i=1}^n U_i(\theta)\right) = \sum_{i=1}^n \mathrm{Var}(U_i(\theta)) = n\,\mathrm{Var}(U_1(\theta)) = \boxed{nI_1(\theta)}.$$

G. Exponential Family (EF)

1. **Exponential family form.**
A (discrete or continuous) statistical model belongs to the exponential family if its pmf/pdf can be written as

$$f(x;\theta) = \exp\{Q(\theta)A(x) + c(x) - k(\theta)\},$$

for suitable functions $A(\cdot)$ (statistic in x), $Q(\cdot)$, $c(\cdot)$, $k(\cdot)$.

2. **Bernoulli as an EF example.**
If $X \sim \mathrm{Ber}(\theta)$, $\theta \in (0,1)$,

$$p_X(x;\theta) = \theta^x (1-\theta)^{1-x} \mathbf{1}_{\{0,1\}}(x).$$

Then

$$p_X(x;\theta) = \exp\{\log p_X(x;\theta)\} = \exp\left\{x \log\theta + (1-x)\log(1-\theta)\right\}$$
$$= \exp\left\{x \log\frac{\theta}{1-\theta} + \log(1-\theta)\right\}.$$

Thus one identification is:

$$A(x) = x, \qquad Q(\theta) = \log\frac{\theta}{1-\theta}, \qquad c(x) = 0, \qquad k(\theta) = -\log(1-\theta).$$

In the most cases $A(x) = x$

3. **Mean identity in EF (regular case) for $A(X)$.**
Assume regularity so that we can differentiate under the integral/sum. From $\int f(x;\theta)\,dx = 1$ we have $\int \partial_\theta f(x;\theta)\,dx = 0$. With

$$f(x;\theta) = \exp\{A(x)Q(\theta) + c(x) - k(\theta)\},$$

we get

$$\frac{\partial}{\partial\theta} f(x;\theta) = \big(A(x)Q'(\theta) - k'(\theta)\big) f(x;\theta).$$

Integrating:

$$0 = \int \big(A(x)Q'(\theta) - k'(\theta)\big) f(x;\theta)\,dx = Q'(\theta)\mathbb{E}[A(X)] - k'(\theta),$$

so

$$\boxed{\mathbb{E}[A(X)] = \frac{k'(\theta)}{Q'(\theta)}}.$$

4. **Variance identity in EF (regular case) for $A(X)$.**
Using again regularity, from $\int f(x;\theta)\,dx = 1$ we have $\int \partial^2_{\theta\theta} f(x;\theta)\,dx = 0$. Differentiate once more:

$$\frac{\partial^2}{\partial\theta^2} f(x;\theta) = \frac{\partial}{\partial\theta}\Big(\big(A(x)Q'(\theta) - k'(\theta)\big) f(x;\theta)\Big) = \big(A(x)Q''(\theta) - k''(\theta)\big) f(x;\theta) + \big(A(x)Q'(\theta) - k'(\theta)\big)\frac{\partial}{\partial\theta} f(x;\theta).$$

Substituting $\partial_\theta f(x;\theta) = \big(A(x)Q'(\theta) - k'(\theta)\big) f(x;\theta)$:

$$\frac{\partial^2}{\partial\theta^2} f(x;\theta) = \Big(\big(A(x)Q''(\theta) - k''(\theta)\big) + \big(A(x)Q'(\theta) - k'(\theta)\big)^2\Big) f(x;\theta).$$

Integrating:

$$0 = \int \frac{\partial^2}{\partial\theta^2} f(x;\theta)\,dx = \int \Big(\big(A(x)Q''(\theta) - k''(\theta)\big) + \big(A(x)Q'(\theta) - k'(\theta)\big)^2\Big) f(x;\theta)\,dx,$$

hence

$$0 = Q''(\theta)\mathbb{E}[A(X)] - k''(\theta) + \mathbb{E}\big[(A(X)Q'(\theta) - k'(\theta))^2\big].$$

Rewrite the squared term:

$$\mathbb{E}\big[(A(X)Q'(\theta) - k'(\theta))^2\big] = (Q'(\theta))^2\,\mathbb{E}\left[\left(A(X) - \frac{k'(\theta)}{Q'(\theta)}\right)^2\right].$$

Using the mean identity $\mathbb{E}[A(X)] = \frac{k'(\theta)}{Q'(\theta)}$, we obtain

$$\mathbb{E}\left[\left(A(X) - \frac{k'(\theta)}{Q'(\theta)}\right)^2\right] = \mathrm{Var}(A(X)),$$

so

$$0 = Q''(\theta)\mathbb{E}[A(X)] - k''(\theta) + (Q'(\theta))^2 \mathrm{Var}(A(X)).$$

Therefore

$$\mathrm{Var}(A(X)) = \frac{k''(\theta) - Q''(\theta)\mathbb{E}[A(X)]}{(Q'(\theta))^2} = \frac{k''(\theta)}{(Q'(\theta))^2} - \frac{Q''(\theta)\,k'(\theta)}{(Q'(\theta))^3},$$

i.e.

$$\boxed{\mathrm{Var}(A(X)) = -\frac{Q''(\theta)\,k'(\theta)}{(Q'(\theta))^3} + \frac{k''(\theta)}{(Q'(\theta))^2}}.$$

5. **Natural parameterization and natural EF form.**
Define the *natural parameter* $\eta := Q(\theta)$. Writing the model in terms of η,

$$f(x;\eta) = \exp\{\eta A(x) + c(x) - \tilde{k}(\eta)\},$$

where $\tilde{k}(\eta) = k(\theta(\eta))$. This is the *natural exponential family* form. In the special case $Q(\theta) = \theta$ (so $\eta = \theta$), the identities simplify to

$$\mathbb{E}[A(X)] = \tilde{k}'(\eta), \qquad \mathrm{Var}(A(X)) = \tilde{k}''(\eta).$$

6. **MLE in the natural parameterization (likelihood equation).**
For an i.i.d. sample $X_1, \ldots, X_n$ from the natural EF,

$$f(x;\eta) = \exp\{\eta A(x) + c(x) - \tilde{k}(\eta)\},$$

the likelihood is

$$\mathcal{L}(\eta;\underline{x}) = \prod_{i=1}^{n} \exp\{\eta A(x_i) + c(x_i) - \tilde{k}(\eta)\} = \exp\left\{\eta \sum_{i=1}^{n} A(x_i) + \sum_{i=1}^{n} c(x_i) - n\tilde{k}(\eta)\right\}.$$

Hence the log-likelihood derivative is

$$\frac{d}{d\eta} \log \mathcal{L}(\eta;\underline{x}) = \sum_{i=1}^{n} A(x_i) - n\tilde{k}'(\eta).$$

The MLE $\hat{\eta}$ solves the likelihood equation

$$\boxed{\tilde{k}'(\hat{\eta}) = \frac{1}{n}\sum_{i=1}^{n} A(x_i)}.$$

7. **EF structure, MLE dependence on $\sum A(X_i)$, and sufficiency.(inutile tranne ultimo rigo)**
From the likelihood equation, $\hat{\eta}$ (hence $\hat{\theta}$) is a function of $\frac{1}{n}\sum_{i=1}^{n} A(x_i)$ (equivalently $\sum_{i=1}^{n} A(x_i)$). Moreover, the joint density/mass is

$$f_{\underline{X}}(\underline{x};\eta) = \exp\left\{\eta \sum_{i=1}^{n} A(x_i) - n\tilde{k}(\eta)\right\} \cdot \exp\left\{\sum_{i=1}^{n} c(x_i)\right\}.$$

This is a Fisher–Neyman factorization with

$$g\left(\sum_{i=1}^{n} A(x_i);\eta\right) = \exp\left\{\eta \sum_{i=1}^{n} A(x_i) - n\tilde{k}(\eta)\right\}, \qquad h(\underline{x}) = \exp\left\{\sum_{i=1}^{n} c(x_i)\right\}.$$

Therefore,

$$\boxed{T(\underline{X}) = \sum_{i=1}^{n} A(X_i) \text{ is sufficient for } \eta \text{ (and for } \theta).}$$

So, in regular exponential families, the MLE depends on the sample only through a sufficient statistic.

H. Sufficiency and Factorization

1. **Definition of sufficient statistic via conditional distribution.**
   Let $\underline{X} = (X_1, \ldots, X_n)$ be a random sample from a model F_θ (discrete or continuous) and let $T_n = T(\underline{X}) \in \mathbb{R}^m$ $(m < n)$.
   T_n is sufficient for θ if the conditional distribution of $\underline{X}$ given T_n does not depend on θ. In the discrete case, for $(x_1, \ldots, x_n) = \underline{x}$ and t,
   $$\mathbb{P}\big(\underline{X} = \underline{x} \mid T_n = t\big) = \frac{\mathbb{P}\big(\underline{X} = \underline{x} \cap \{T_n = t\}\big)}{\mathbb{P}(T_n = t)}.$$

2. **Bernoulli example: $T_n = \sum_{i=1}^n X_i$ is sufficient for θ.** Assume $X_1, \ldots, X_n \overset{iid}{\sim} \mathrm{Ber}(\theta)$ and let $\underline{x} = (x_1, \ldots, x_n) \in \{0,1\}^n$.
   (a) **Conditional distribution given the total number of successes.** The joint pmf is
   $$p_{\underline{X}}(\underline{x}; \theta) = \prod_{i=1}^n \theta^{x_i}(1-\theta)^{1-x_i} = \theta^{\sum_{i=1}^n x_i}(1-\theta)^{n-\sum_{i=1}^n x_i}.$$
   Define the statistic
   $$T_n = \sum_{i=1}^n X_i \in \{0, \ldots, n\}, \qquad \text{so that} \quad T_n \sim \mathrm{Bin}(n, \theta).$$
   For any $\underline{x} \in \{0,1\}^n$ such that $\sum_{i=1}^n x_i = t$,
   $$\mathbb{P}(\underline{X} = \underline{x} \mid T_n = t) = \frac{\mathbb{P}(\underline{X} = \underline{x})}{\mathbb{P}(T_n = t)} = \frac{\theta^t(1-\theta)^{n-t}}{\binom{n}{t}\theta^t(1-\theta)^{n-t}} = \frac{1}{\binom{n}{t}},$$
   which does not depend on θ. Hence T_n is sufficient for θ.
   (b) **Partition viewpoint (uniformity over each fiber).** For each $t \in \{0, \ldots, n\}$ define the level set (fiber)
   $$I_t := \Big\{\underline{x} \in \{0,1\}^n : \sum_{i=1}^n x_i = t\Big\}.$$
   Then $\{0,1\}^n = \bigcup_{t=0}^n I_t$ and the sets I_t are pairwise disjoint. Moreover,
   $$|I_t| = \binom{n}{t},$$
   since choosing $\underline{x} \in I_t$ amounts to choosing which t coordinates equal 1. Therefore, conditional on $\{T_n = t\}$, the random vector $\underline{X}$ is *uniformly distributed* over I_t.

3. **Discrete case reminder: conditioning on $\{T(\underline{X}) = t(\underline{x})\}$.**
   (Only for discrete models.) Fix an observed sample point $\underline{x}$ and set $t(\underline{x}) := T(\underline{x})$. Then
   $$\mathbb{P}\big(\underline{X} = \underline{x} \,\big|\, T(\underline{X}) = t(\underline{x})\big) = \frac{\mathbb{P}\big(\{\underline{X} = \underline{x}\} \cap \{T(\underline{X}) = t(\underline{x})\}\big)}{\mathbb{P}\big(T(\underline{X}) = t(\underline{x})\big)}.$$
   Since $\{\underline{X} = \underline{x}\} \subseteq \{T(\underline{X}) = t(\underline{x})\}$, it follows that
   $$\mathbb{P}\big(\underline{X} = \underline{x} \,\big|\, T(\underline{X}) = t(\underline{x})\big) = \frac{\mathbb{P}(\underline{X} = \underline{x})}{\mathbb{P}\big(T(\underline{X}) = t(\underline{x})\big)}.$$

4. **Fisher criterion theorem for sufficiency.**
   Let $p_{\underline{X}}(\underline{x}; \theta)$ be the joint density/pmf of $\underline{X}$, and let $q_T(t; \theta)$ be the density/pmf of $T(\underline{X})$. Then $T(\underline{X})$ is sufficient for θ if, for every $\underline{x}$ in the sample space,
   $$\frac{p_{\underline{X}}(\underline{x}; \theta)}{q_T\big(T(\underline{x}); \theta\big)} \quad \text{is independent of } \theta.$$
   In the continuous case, this criterion is more delicate because densities are only defined up to sets of measure zero and the ratio can be ill-defined where $q_T\big(T(x); \theta\big) = 0$, so the condition is typically understood "almost everywhere" rather than pointwise.

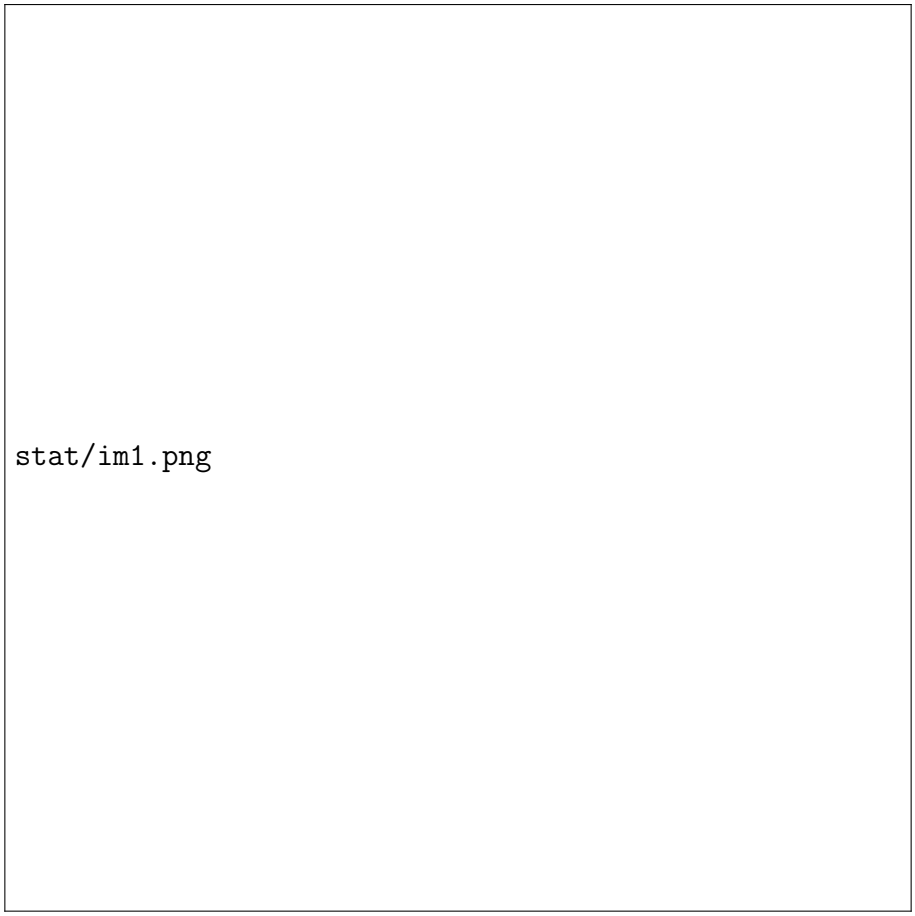


Figure 1: Partition of $\{0,1\}^n$ into the fibers $I_t = \{\underline{x} : \sum_i x_i = t\}$.

5. **Fisher–Neyman factorization corollary** $p(\underline{x};\theta) = g(T(\underline{x});\theta)h(\underline{x})$.
Corollary (Factorization / Savage). A statistic $T(\underline{X})$ is sufficient for θ if and only if there exist two non-negative functions $g(\cdot)$ and $h(\cdot)$ such that for all $\underline{x}$ and θ,

$$p_{\underline{X}}(\underline{x};\theta) = g\big(T(\underline{x});\theta\big)\, h(\underline{x}).$$

Bernoulli illustration.

$$p_{\underline{X}}(\underline{x};\theta) = \theta^{\sum x_i}(1-\theta)^{n-\sum x_i} = g(t;\theta)\, h(\underline{x}), \quad g(t;\theta) = \theta^t(1-\theta)^{n-t}, \quad h(\underline{x}) = 1,$$

so $T(\underline{x}) = \sum x_i$ is sufficient.

Proof. **Poof structure of the factorization theorem (both directions.**
($\Rightarrow$) Sufficiency $\implies$ factorization (discrete sketch).
Assume T is sufficient. Define

$$g(t;\theta) = \mathbb{P}\big(T(\underline{X}) = t\big), \qquad h(\underline{x}) = \mathbb{P}\big(\underline{X} = \underline{x} \mid T(\underline{X}) = t(\underline{x})\big), \qquad t(\underline{x}) := T(\underline{x}).$$

Then, since $\{\underline{X} = \underline{x}\} \subseteq \{T(\underline{X}) = t(\underline{x})\}$,

$$p_{\underline{X}}(\underline{x};\theta) = \mathbb{P}(\underline{X} = \underline{x}) = \mathbb{P}\big(T(\underline{X}) = t(\underline{x})\big)\, \mathbb{P}\big(\underline{X} = \underline{x} \mid T(\underline{X}) = t(\underline{x})\big) = g\big(t(\underline{x});\theta\big)\, h(\underline{x}).$$

($\Leftarrow$) Factorization $\implies$ sufficiency (discrete sketch).
Assume $p_{\underline{X}}(\underline{x};\theta) = g(T(\underline{x});\theta)h(\underline{x})$ and let $q_T(\cdot;\theta)$ be the pmf of $T(\underline{X})$. For fixed $\underline{x}$, define

$$A_{T(\underline{x})} = \{\underline{y} :\ T(\underline{y}) = T(\underline{x})\}.$$

Then

$$q_T\big(T(\underline{x});\theta\big) = \sum_{\underline{y} \in A_{T(\underline{x})}} p_{\underline{X}}(\underline{y};\theta) = \sum_{\underline{y} \in A_{T(\underline{x})}} g(T(\underline{y});\theta)\, h(\underline{y}) = g\big(T(\underline{x});\theta\big) \sum_{\underline{y} \in A_{T(\underline{x})}} h(\underline{y}).$$

Hence

$$\frac{p_{\underline{X}}(\underline{x};\theta)}{q_T(T(\underline{x});\theta)} = \frac{g(T(\underline{x});\theta)h(\underline{x})}{g(T(\underline{x});\theta)\sum_{\underline{y}\in A_{T(\underline{x})}} h(\underline{y})} = \frac{h(\underline{x})}{\sum_{\underline{y}\in A_{T(\underline{x})}} h(\underline{y})},$$

which is independent of θ. By Fisher's ratio criterion, T is sufficient. □

6. **Applications / remarks.**

(a) **Gaussian mean unknown, variance known: $\bar{X}$ sufficient for μ.**

Let $X_1, \ldots, X_n \overset{iid}{\sim} N(\mu, \sigma^2)$ with σ^2 known. The joint density is

$$f_{\underline{X}}(\underline{x}; \mu, \sigma^2) = (2\pi\sigma^2)^{-n/2} \exp\left\{ -\frac{1}{2\sigma^2} \sum_{i=1}^{n} (x_i - \mu)^2 \right\}.$$

Let $T_n = \bar{X} = \frac{1}{n}\sum_{i=1}^{n} x_i =: t$, with

$$f_{T_n}(t; \mu, \sigma^2) = \left(2\pi \frac{\sigma^2}{n}\right)^{-1/2} \exp\left\{ -\frac{n}{2\sigma^2}(t-\mu)^2 \right\}.$$

Using

$$\sum_{i=1}^{n} (x_i - \mu)^2 = \sum_{i=1}^{n} (x_i - t + t - \mu)^2 = \sum_{i=1}^{n} (x_i - t)^2 + n(t-\mu)^2 \quad \Big(\text{since } \sum_{i=1}^{n} (x_i - t) = 0\Big),$$

we can rewrite the joint density as

$$f_{\underline{X}}(\underline{x}; \mu, \sigma^2) = (2\pi\sigma^2)^{-n/2} \exp\left\{ -\frac{1}{2\sigma^2} \sum_{i=1}^{n} (x_i - t)^2 \right\} \exp\left\{ -\frac{n}{2\sigma^2}(t-\mu)^2 \right\}.$$

Therefore

$$\frac{f_{\underline{X}}(\underline{x}; \mu, \sigma^2)}{f_{T_n}(t; \mu, \sigma^2)} = \frac{(2\pi\sigma^2)^{-n/2}}{\big(2\pi\sigma^2/n\big)^{-1/2}} \exp\left\{ -\frac{1}{2\sigma^2} \sum_{i=1}^{n} (x_i - t)^2 \right\},$$

which does not depend on μ. Hence, by the Neyman–Fisher factorization criterion (equivalently by the ratio criterion), $\bar{X}$ is sufficient for μ.

Equivalently, factorize:

$$f_{\underline{X}}(\underline{x}; \mu, \sigma^2) = (2\pi\sigma^2)^{-n/2} \exp\left\{ -\frac{1}{2\sigma^2} \sum x_i^2 \right\} \exp\left\{ \frac{\mu}{\sigma^2} \sum x_i - \frac{n\mu^2}{2\sigma^2} \right\} = h(\underline{x})\, g\big(\textstyle\sum x_i; \mu\big),$$

hence $\bar{X}$ (equivalently $\sum x_i$) is sufficient.

(b) **Exponential family: $\sum A(X_i)$ sufficient.**

For a (natural) exponential family

$$f_X(x; \mu) = \exp\{A(x)\mu + c(x) - \tilde{k}(\mu)\},$$

the joint density is

$$f_{\underline{X}}(\underline{x}; \mu) = \prod_{i=1}^{n} \exp\{A(x_i)\mu + c(x_i) - \tilde{k}(\mu)\} = \exp\left\{ \mu \sum_{i=1}^{n} A(x_i) - n\tilde{k}(\mu) \right\} \exp\left\{ \sum_{i=1}^{n} c(x_i) \right\}.$$

By factorization, $\sum_{i=1}^{n} A(X_i)$ is sufficient for μ.

Remark (as in notes): in regular models, MLEs are functions of a sufficient statistic (here, of $\sum A(X_i)$).

(c) **Trivial sufficient statistic (full sample).**

For any model F_θ, the full sample $\underline{X} = (X_1, \ldots, X_n)$ is always sufficient:

$$p_{\underline{X}}(\underline{x}; \theta) = g(\underline{x}; \theta)\, h(\underline{x}), \qquad g(\underline{x}; \theta) = p_{\underline{X}}(\underline{x}; \theta), \quad h(\underline{x}) = 1.$$

(d) **Bijection invariance of sufficiency.**

If $T(\underline{x})$ is sufficient for θ and $r(\cdot)$ is a bijection, then $T^*(\underline{x}) = r(T(\underline{x}))$ is sufficient. Indeed, if

$$p_{\underline{X}}(\underline{x}; \theta) = g(T(\underline{x}); \theta)\, h(\underline{x}),$$

then, writing $T(\underline{x}) = r^{-1}(T^*(\underline{x}))$ and defining $g^*(t; \theta) = g(r^{-1}(t); \theta)$,

$$p_{\underline{X}}(\underline{x}; \theta) = g^*(T^*(\underline{x}); \theta)\, h(\underline{x}),$$

so T^* is sufficient by factorization.

(e) **Counter-example reference: Beta model not in EF (as mentioned).**
Example in notes: $X \sim \text{Beta}(\alpha, 1)$ with $\alpha > 0$,

$$f_X(x;\alpha) \propto x^{\alpha-1}\mathbf{1}_{(0,1)}(x).$$

(Referenced as a counter-example: this model is not in the exponential family in the sense discussed in the notes; the notes mention finding the MLE α^* and searching for a sufficient statistic via factorization.)

(f) **Uniform $U(a-b, a+b)$: $(\min X_i, \max X_i)$ sufficient for (a, b).**
If $X_1, \ldots, X_n \overset{iid}{\sim} U(a-b, a+b)$ with $a, b > 0$, the likelihood is

$$\mathcal{L}(a, b; \underline{x}) = \left(\frac{1}{2b}\right)^n \prod_{i=1}^{n} \mathbf{1}_{(a-b,a+b)}(x_i).$$

The indicator product equals 1 iff all observations lie in $(a-b, a+b)$, i.e.

$$\max_i x_i \le a+b \quad \text{and} \quad \min_i x_i \ge a-b.$$

Therefore the likelihood depends on the sample only through $(X_{(1)}, X_{(n)}) = (\min_i X_i, \max_i X_i)$, and by factorization $(\min X_i, \max X_i)$ is sufficient for (a, b).

I. Minimal Sufficiency

1. **Definition via refinement of partitions / functional dependence on any sufficient statistic.**
   (a) A sufficient statistic $T(\underline{x})$ is called *minimal sufficient* if for any other sufficient statistic $T'(\underline{x})$,

   $$T(\underline{x}) \text{ is a function of } T'(\underline{x}).$$

   Equivalently,

   $$T'(\underline{x}) = T'(\underline{y}) \implies T(\underline{x}) = T(\underline{y}).$$

   (b) **Partition language.** Let $\{A_t : t \in \mathcal{T}\}$ be the partition of the sample space induced by T:

   $$A_t = \{\underline{x} : T(\underline{x}) = t\},$$

   and let $\{B_{t'} : t' \in \mathcal{T}'\}$ be the partition induced by T':

   $$B_{t'} = \{\underline{x} : T'(\underline{x}) = t'\}.$$

   Then T is a function of T' if and only if every block $B_{t'}$ is contained in some block A_t (i.e. the partition induced by T' is a refinement of the partition induced by T).
2. **Likelihood ratio criterion characterization (as used).**
   A key characterization (used to certify minimal sufficiency) is based on the joint density/mass $f_{\underline{X}}(\underline{x}; \theta)$:

   $$\frac{f_{\underline{X}}(\underline{x};\theta)}{f_{\underline{X}}(\underline{y};\theta)} \text{ is independent of } \theta \quad \iff \quad T(\underline{x}) = T(\underline{y}).$$

   Intuitively: two samples $\underline{x}, \underline{y}$ are in the same equivalence class (same value of T) exactly when the likelihood ratio does not depend on the parameter.
3. **Lehmann–Scheffé theorem for minimal sufficiency and its proof structure (as in notes).**
   (a) **Theorem (Lehmann–Scheffé).** Let $f_{\underline{X}}(\underline{x}; \theta)$ be the joint density or mass function of $\underline{X} = (X_1, \ldots, X_n)$ from F_θ. Assume there exists a function $T(\underline{x})$ such that for any $\underline{x}, \underline{y}$ in the sample space,

   $$\frac{f_{\underline{X}}(\underline{x};\theta)}{f_{\underline{X}}(\underline{y};\theta)} \text{ is independent of } \theta \quad \iff \quad T(\underline{x}) = T(\underline{y}).$$

   Then $T(\underline{x})$ is *minimal sufficient* for θ.

(b) **Proof structure (as in notes; discrete or continuous, with $f_{\underline{X}}(\underline{x};\theta) > 0$ on the sample space).**

i. *Step 1: show T is sufficient.* Let $\{A_t\}$ be the partition induced by T with

$$A_t = \{\underline{x} : T(\underline{x}) = t\}.$$

For each A_t fix one representative point $\underline{x}_t \in A_t$. For any $\underline{x}$, the representative in the same block is $\underline{x}_{T(\underline{x})}$ and

$$T(\underline{x}) = T(\underline{x}_{T(\underline{x})}).$$

By the assumption,

$$\frac{f_{\underline{X}}(\underline{x};\theta)}{f_{\underline{X}}(\underline{x}_{T(\underline{x})};\theta)} \text{ is independent of } \theta.$$

Define

$$h(\underline{x}) = \frac{f_{\underline{X}}(\underline{x};\theta)}{f_{\underline{X}}(\underline{x}_{T(\underline{x})};\theta)}, \qquad g(t;\theta) = f_{\underline{X}}(\underline{x}_t;\theta).$$

Then

$$f_{\underline{X}}(\underline{x};\theta) = g(T(\underline{x});\theta)\, h(\underline{x}),$$

so by Fisher–Neyman factorization (Savage corollary in the notes), T is sufficient.

ii. *Step 2: show T is minimal.* Let $T'(\underline{x})$ be any other sufficient statistic. By factorization,

$$f_{\underline{X}}(\underline{x};\theta) = g'(T'(\underline{x});\theta)\, h'(\underline{x}).$$

If $T'(\underline{x}) = T'(\underline{y})$, then

$$\frac{f_{\underline{X}}(\underline{x};\theta)}{f_{\underline{X}}(\underline{y};\theta)} = \frac{g'(T'(\underline{x});\theta)h'(\underline{x})}{g'(T'(\underline{y});\theta)h'(\underline{y})} = \frac{h'(\underline{x})}{h'(\underline{y})},$$

which is independent of θ. By the defining assumption of the theorem, this implies

$$T(\underline{x}) = T(\underline{y}).$$

Hence T is a function of T', i.e. T is minimal sufficient.

4. **Examples referenced: Bernoulli and Gaussian.**

(a) **Bernoulli model.** Let $X_1, \ldots, X_n \overset{iid}{\sim} \mathrm{Ber}(\theta)$, $\theta \in (0,1)$. The joint pmf is

$$f_{\underline{X}}(\underline{x};\theta) = \prod_{i=1}^{n} \theta^{x_i}(1-\theta)^{1-x_i} = \theta^{\sum_{i=1}^n x_i}(1-\theta)^{n-\sum_{i=1}^n x_i}, \qquad \underline{x} \in \{0,1\}^n.$$

For two samples $\underline{x}, \underline{y} \in \{0,1\}^n$,

$$\frac{f_{\underline{X}}(\underline{x};\theta)}{f_{\underline{X}}(\underline{y};\theta)} = \left(\frac{\theta}{1-\theta}\right)^{\sum x_i - \sum y_i}.$$

This ratio is independent of θ if and only if $\sum_{i=1}^n x_i = \sum_{i=1}^n y_i$. Therefore, by Lehmann–Scheffé,

$$T(\underline{X}) = \sum_{i=1}^{n} X_i$$

is minimal sufficient for θ (and any bijection of it, e.g. $\bar{X}_n = \frac{1}{n}\sum X_i$, is also sufficient; in particular $\sum X_i$ and $\frac{1}{n}\sum X_i$ generate the same partition).

(b) **Gaussian model (unknown mean, known variance).** Let $X_1, \ldots, X_n \overset{iid}{\sim} \mathcal{N}(\mu, \sigma^2)$ with σ^2 known. The joint density is

$$f_{\underline{X}}(\underline{x};\mu) = (2\pi\sigma^2)^{-n/2} \exp\left\{-\frac{1}{2\sigma^2}\sum_{i=1}^{n}(x_i-\mu)^2\right\}.$$

Expand

$$\sum_{i=1}^{n}(x_i-\mu)^2=\sum_{i=1}^{n}x_i^2-2\mu\sum_{i=1}^{n}x_i+n\mu^2.$$

Hence, for $\underline{x},\underline{y}$,

$$\frac{f_{\underline{X}}(\underline{x};\mu)}{f_{\underline{X}}(\underline{y};\mu)}=\exp\Big\{-\frac{1}{2\sigma^2}\Big(\sum x_i^2-\sum y_i^2\Big)\Big\}\cdot\exp\Big\{\frac{\mu}{\sigma^2}\Big(\sum x_i-\sum y_i\Big)\Big\}.$$

This ratio is independent of μ if and only if $\sum_{i=1}^n x_i=\sum_{i=1}^n y_i$ (equivalently $\bar{x}=\bar{y}$). Therefore, by Lehmann–Scheffé,

$$T(\underline{X})=\sum_{i=1}^{n}X_i\quad\text{(equivalently }\bar{X}_n=\tfrac{1}{n}\sum_{i=1}^{n}X_i\text{)}$$

is minimal sufficient for μ.

J. Properties of Estimators

1. **Finite-sample vs asymptotic properties.**
   Let $\underline{X}=(X_1,\dots,X_n)$ be a random sample from $X\sim F_\theta$, $\theta\in\Theta$ (regular or non-regular), and let $T_n=T(X_1,\dots,X_n)$ be an estimator of θ (i.e. a random variable).
   - *Finite-sample properties*: statements that hold for a fixed $n\geq 1$ (e.g. exact unbiasedness, exact variance, exact distribution).
   - *Asymptotic properties*: statements about the behavior as $n\to+\infty$ (e.g. asymptotic correctness, consistency, asymptotic normality).
2. **Unbiasedness and bias $b_n(\theta)$.**
   T_n is *unbiased* (or *correct*) for θ if
   $$\mathbb{E}[T_n]=\theta.$$
   The *bias* of T_n is
   $$b_n(\theta):=\mathbb{E}[T_n]-\theta.$$
3. **Unbiasedness of the sample mean when $\mathbb{E}[X]=\theta$.**
   Assume $\mathbb{E}[X]=\theta$ (e.g. Bernoulli, Poisson, Gaussian in the parametrization where the mean equals θ). Define the sample mean
   $$\bar{X}_n=\frac{1}{n}\sum_{i=1}^{n}X_i.$$
   Then
   $$\mathbb{E}[\bar{X}_n]=\mathbb{E}\left[\frac{1}{n}\sum_{i=1}^{n}X_i\right]=\frac{1}{n}\sum_{i=1}^{n}\mathbb{E}[X_i]=\frac{1}{n}\sum_{i=1}^{n}\theta=\theta,$$
   hence the sample mean is unbiased for θ whenever $\mathbb{E}[X]=\theta$.
4. **Bias of the uncorrected sample variance $\tilde{S}_n^2$; corrected variance S_n^2 unbiased.**
   Let $\theta=(\mu,\sigma^2)$ with $\mu=\mathbb{E}[X]$ and $\sigma^2=\mathrm{Var}(X)$ (e.g. Gaussian). Consider the *uncorrected* sample variance as an estimator of σ^2:
   $$\tilde{S}_n^2=\frac{1}{n}\sum_{i=1}^{n}(X_i-\bar{X}_n)^2.$$
   Compute:
   $$\mathbb{E}[\tilde{S}_n^2]=\mathbb{E}\left[\frac{1}{n}\sum_{i=1}^{n}(X_i-\bar{X}_n)^2\right]=\mathbb{E}\left[\frac{1}{n}\sum_{i=1}^{n}X_i^2-\bar{X}_n^2\right]=\frac{1}{n}\sum_{i=1}^{n}\mathbb{E}[X_i^2]-\mathbb{E}[\bar{X}_n^2].$$

Now $\mathbb{E}[X_i^2] = \sigma^2 + \mu^2$, so $\frac{1}{n}\sum_{i=1}^n \mathbb{E}[X_i^2] = \sigma^2 + \mu^2$. Also $\mathbb{E}[\bar{X}_n] = \mu$ and $\mathrm{Var}(\bar{X}_n) = \sigma^2/n$, hence

$$\mathbb{E}[\bar{X}_n^2] = \mathrm{Var}(\bar{X}_n) + (\mathbb{E}[\bar{X}_n])^2 = \frac{\sigma^2}{n} + \mu^2.$$

Therefore,

$$\mathbb{E}[\tilde{S}_n^2] = (\sigma^2 + \mu^2) - \left(\frac{\sigma^2}{n} + \mu^2\right) = \sigma^2\left(1 - \frac{1}{n}\right) = \sigma^2\frac{n-1}{n}.$$

So $\tilde{S}_n^2$ is biased downward for σ^2, but becomes asymptotically correct since $\frac{n-1}{n} \to 1$. The *corrected* sample variance

$$S_n^2 = \frac{1}{n-1}\sum_{i=1}^n (X_i - \bar{X}_n)^2$$

is unbiased for σ^2.

5. **Mean squared error (MSE) decomposition:** $\mathrm{MSE} = \mathrm{Var} + b^2$**.**
For a generic estimator T_n of θ, consider the distance $|T_n - \theta|$. By Chebyshev's inequality, for any $k > 0$,

$$\mathbb{P}(|T_n - \theta| < k) \geq 1 - \frac{\mathbb{E}[(T_n - \theta)^2]}{k^2}.$$

This motivates the *mean squared error*:

$$\mathrm{MSE}(T_n) := \mathbb{E}[(T_n - \theta)^2].$$

Decompose:

$$\mathrm{MSE}(T_n) = \mathbb{E}\big[(T_n - \mathbb{E}[T_n] + \mathbb{E}[T_n] - \theta)^2\big] = \mathrm{Var}(T_n) + b_n^2(\theta),$$

where $b_n(\theta) = \mathbb{E}[T_n] - \theta$ (the cross term is 0). If T_n is unbiased, then $\mathrm{MSE}(T_n) = \mathrm{Var}(T_n)$.

6. **Relative efficiency via MSE comparison.**
If T_{1n} and T_{2n} are two estimators of θ, then T_{1n} is *more efficient* than T_{2n} if

$$\mathrm{MSE}(T_{1n}) < \mathrm{MSE}(T_{2n}).$$

In particular, if both are unbiased, this reduces to comparing variances:

$$\mathbb{E}[T_{1n}] = \mathbb{E}[T_{2n}] = \theta \quad \Rightarrow \quad \mathrm{MSE}(T_{in}) = \mathrm{Var}(T_{in}), \quad i = 1, 2,$$

so smaller variance means higher efficiency.

K. Cramér–Rao Bound and Efficient Estimation

1. **Cramér–Rao bound (regularity).**
Theorem. Consider $\mathbf{X} = (X_1, \ldots, X_n)$ a random sample from a *regular* model $\{f(\cdot;\theta) : \theta \in \Theta\}$. Then, for any unbiased (correct) estimator T_n of θ,

$$\boxed{\mathrm{Var}_\theta(T_n) \;\geq\; \frac{1}{I_n(\theta)}}$$

where $I_n(\theta)$ is the Fisher information of the sample:

$$I_n(\theta) = \mathrm{Var}_\theta\big(V_n'(\theta)\big), \qquad V_n'(\theta) = \frac{\partial}{\partial\theta}\log L(\theta;\mathbf{X}).$$

Proof. Let T_n be any estimator of θ (biased or unbiased) and define its bias

$$b_n(\theta) = \mathbb{E}_\theta[T_n] - \theta \qquad \Longrightarrow \qquad \mathbb{E}_\theta[T_n] = \theta + b_n(\theta).$$

Under regularity,

$$\mathbb{E}_\theta\big[V_n'(\theta)\big] = 0, \qquad I_n(\theta) = \mathbb{E}_\theta\big[(V_n'(\theta))^2\big] = -\mathbb{E}_\theta\big[V_n''(\theta)\big].$$

Step 1: compute $\mathrm{Cov}_\theta(T_n, V_n'(\theta))$. Since $\mathbb{E}_\theta[V_n'(\theta)] = 0$,

$$\mathrm{Cov}_\theta(T_n, V_n'(\theta)) = \mathbb{E}_\theta[T_n\, V_n'(\theta)] - 0$$

Writing the expectation as an integral (with joint density $f_{\mathbf{X}}(\mathbf{x};\theta)$),

$$\mathbb{E}_\theta[T_n\, V_n'(\theta)] = \int_{\mathbb{R}^n} T_n(\mathbf{x})\, \frac{\partial}{\partial\theta} \log f_{\mathbf{X}}(\mathbf{x};\theta)\, f_{\mathbf{X}}(\mathbf{x};\theta)\, d\mathbf{x}.$$

Use $\frac{\partial}{\partial\theta}\log f \cdot f = \frac{\partial}{\partial\theta} f$:

$$\mathrm{Cov}_\theta(T_n, V_n'(\theta)) = \int_{\mathbb{R}^n} T_n(\mathbf{x})\, \frac{\partial}{\partial\theta} f_{\mathbf{X}}(\mathbf{x};\theta)\, d\mathbf{x} = \frac{d}{d\theta} \int_{\mathbb{R}^n} T_n(\mathbf{x})\, f_{\mathbf{X}}(\mathbf{x};\theta)\, d\mathbf{x}.$$

Therefore,

$$\mathrm{Cov}_\theta(T_n, V_n'(\theta)) = \frac{d}{d\theta}\mathbb{E}_\theta[T_n] = \frac{d}{d\theta}\big(\theta + b_n(\theta)\big) = 1 + b_n'(\theta).$$

Step 2: apply Cauchy–Schwarz.

$$\mathrm{Cov}_\theta(T_n, V_n'(\theta))^2 \le \mathrm{Var}_\theta(T_n)\, \mathrm{Var}_\theta(V_n'(\theta)) = \mathrm{Var}_\theta(T_n)\, I_n(\theta).$$

Hence

$$\mathrm{Var}_\theta(T_n) \ge \frac{(1 + b_n'(\theta))^2}{I_n(\theta)}.$$

If T_n is unbiased for θ, then $b_n(\theta) \equiv 0$, so

$$\mathrm{Var}_\theta(T_n) \ge \frac{1}{I_n(\theta)}.$$

□

2. **General (biased) CRLB and induced MSE lower bound.**
Let T_n be any estimator of θ with bias $b_n(\theta) = \mathbb{E}_\theta[T_n] - \theta$. Then

$$\boxed{\mathrm{Var}_\theta(T_n) \;\ge\; \frac{(1 + b_n'(\theta))^2}{I_n(\theta)}}$$

and, since $\mathrm{MSE}_\theta(T_n) = \mathrm{Var}_\theta(T_n) + b_n(\theta)^2$,

$$\boxed{\mathrm{MSE}_\theta(T_n) \;\ge\; \frac{(1 + b_n'(\theta))^2}{I_n(\theta)} + b_n(\theta)^2}.$$

3. **Prop efficiency.**
   (a) **Definition (attaining CRLB).**
   An unbiased estimator T_n of θ is **efficient** (finite-sample efficient) if it attains the CRLB:

$$\mathbb{E}_\theta[T_n] = \theta, \qquad \mathrm{Var}_\theta(T_n) = \frac{1}{I_n(\theta)}.$$

   (b) **Uniqueness of efficient unbiased estimators.**
   Proposition. In a regular model, if T_n is unbiased and efficient for θ, then it is unique.
   Argument (as in notes). Suppose T_{1n} and T_{2n} satisfy

$$\mathbb{E}[T_{1n}] = \mathbb{E}[T_{2n}] = \theta, \qquad \mathrm{Var}(T_{1n}) = \mathrm{Var}(T_{2n}) = \frac{1}{I_n(\theta)} =: v.$$

Consider $T_n = \frac{T_{1n}+T_{2n}}{2}$, then $\mathbb{E}[T_n] = \theta$ and

$$\mathrm{Var}(T_n) = \mathrm{Var}\left(\frac{T_{1n}+T_{2n}}{2}\right) = \frac{1}{4}\Big(\mathrm{Var}(T_{1n}) + \mathrm{Var}(T_{2n}) + 2\mathrm{Cov}(T_{1n}, T_{2n})\Big).$$

Let $\rho = \mathrm{Corr}(T_{1n}, T_{2n}) \in [-1, 1]$; since $\mathrm{Cov} = \rho\sqrt{\mathrm{VarVar}} = \rho v$,

$$\mathrm{Var}(T_n) = \frac{1}{4}(2v + 2\rho v) = \frac{1}{2}v(1+\rho) \le v.$$

But v is the minimum possible variance among unbiased estimators (CRLB), so we must have $\mathrm{Var}(T_n) = v$, hence $\rho = 1$. Thus $T_{1n} = a + bT_{2n}$. Taking expectations:

$$\theta = a + b\theta \;\Rightarrow\; a = 0,\; b = 1 \;\Rightarrow\; T_{1n} = T_{2n}.$$

Remark: uniqueness is for *correct* (unbiased) efficient estimators.

(c) **Existence characterization:** $V_n'(\theta) = I_n(\theta)\,[T_n - \theta]$**.**

Proposition. Consider a regular model. An estimator T_n is **unbiased and efficient** for θ *iff*

$$\boxed{V_n'(\theta) = \frac{\partial}{\partial\theta}\log L(\theta; \mathbf{X}) = I_n(\theta)\,[T_n - \theta]}$$

($\Rightarrow$) If the identity holds. From

$$T_n = \frac{V_n'(\theta)}{I_n(\theta)} + \theta$$

we get

$$\mathbb{E}[T_n] = \frac{\mathbb{E}[V_n'(\theta)]}{I_n(\theta)} + \theta = \theta, \qquad \mathrm{Var}(T_n) = \frac{1}{I_n(\theta)^2}\mathrm{Var}(V_n'(\theta)) = \frac{1}{I_n(\theta)}.$$

($\Leftarrow$) If T_n is unbiased and efficient. In the CR proof, Cauchy–Schwarz becomes equality, hence T_n and $V_n'(\theta)$ are perfectly correlated, so

$$V_n'(\theta) = a + bT_n.$$

Using $\mathbb{E}[V_n'(\theta)] = 0$ and $\mathbb{E}[T_n] = \theta$ gives $a = -b\theta$, hence

$$V_n'(\theta) = b(T_n - \theta).$$

To identify b, multiply both sides by $V_n'(\theta)$ and take expectations:

$$\mathbb{E}\big[(V_n'(\theta))^2\big] = \mathbb{E}\big[V_n'(\theta)\, b(T_n - \theta)\big] = b\,\mathbb{E}\big[T_n V_n'(\theta)\big] - b\theta\,\mathbb{E}\big[V_n'(\theta)\big] = b\,\mathbb{E}\big[T_n V_n'(\theta)\big],$$

since $\mathbb{E}[V_n'(\theta)] = 0$. But $\mathbb{E}[(V_n'(\theta))^2] = I_n(\theta)$, and for unbiased T_n we have

$$\mathbb{E}[T_n V_n'(\theta)] = \mathrm{Cov}(T_n, V_n'(\theta)) = 1 \qquad (+b'(\theta) = 0 \text{ from hp})$$

(from Step 1 in the CR proof with $b_n' \equiv 0$). Therefore

$$I_n(\theta) = b \cdot 1 \quad \Rightarrow \quad b = I_n(\theta),$$

and the characterization follows.

4. **Link with exponential-family structure (as in the notes).**
Let

$$V_n'(\theta) \;:=\; \frac{\partial}{\partial\theta}\log L(\theta; \mathbf{X})$$

denote the score (derivative of the log-likelihood). Assume there exists a statistic $T_n = T_n(\mathbf{X})$ that does not depend on θ such that the score is affine in T_n:

$$V_n'(\theta) = c_0(\theta) + c_1(\theta)\, T_n,$$

where $c_0(\theta)$ and $c_1(\theta)$ are *functions of* θ (they may vary with θ; the linearity is in T_n, not in θ). By the usual regularity conditions,

$$\mathbb{E}_\theta[V_n'(\theta)] = 0.$$

Write the (possibly biased) mean of T_n as

$$\mathbb{E}_\theta[T_n] = \theta + b(\theta).$$

Taking expectations in $V_n'(\theta) = c_0(\theta) + c_1(\theta)T_n$ yields

$$0 = c_0(\theta) + c_1(\theta)\,\mathbb{E}_\theta[T_n] = c_0(\theta) + c_1(\theta)\big(\theta + b(\theta)\big),$$

hence

$$c_0(\theta) = -c_1(\theta)\big(\theta + b(\theta)\big), \qquad \Rightarrow \qquad V_n'(\theta) = c_1(\theta)\Big(T_n - \theta - b(\theta)\Big).$$

Next compute the covariance between T_n and the score. Since $\mathbb{E}_\theta[V_n'(\theta)] = 0$,

$$\mathrm{Cov}_\theta(T_n, V_n'(\theta)) = \mathbb{E}_\theta[T_n V_n'(\theta)].$$

Using $V_n'(\theta) = c_1(\theta)\big(T_n - \mathbb{E}_\theta[T_n]\big)$, we get

$$\mathrm{Cov}_\theta(T_n, V_n'(\theta)) = c_1(\theta)\,\mathbb{E}_\theta\Big[T_n\big(T_n - \mathbb{E}_\theta[T_n]\big)\Big] = c_1(\theta)\ \mathrm{Var}_\theta(T_n).$$

On the other hand, by the standard identity used in the Cramér–Rao proof,

$$\mathrm{Cov}_\theta(T_n, V_n'(\theta)) = \frac{d}{d\theta}\mathbb{E}_\theta[T_n] = \frac{d}{d\theta}\big(\theta + b(\theta)\big) = 1 + b'(\theta).$$

Therefore,

$$c_1(\theta)\ \mathrm{Var}_\theta(T_n) = 1 + b'(\theta).$$

If T_n attains the (possibly biased) Cramér–Rao lower bound,

$$\mathrm{Var}_\theta(T_n) = \frac{(1 + b'(\theta))^2}{I_n(\theta)},$$

then necessarily

$$c_1(\theta) = \frac{I_n(\theta)}{1 + b'(\theta)}, \qquad \text{and} \qquad V_n'(\theta) = \frac{I_n(\theta)}{1 + b'(\theta)}\Big(T_n - \theta - b(\theta)\Big).$$

In particular, in the unbiased efficient case ($b \equiv 0$),

$$V_n'(\theta) = I_n(\theta)\,(T_n - \theta).$$

Finally, integrate the score identity $V_n'(\theta) = \partial_\theta \log L(\theta; \mathbf{x})$ with respect to θ:

$$\log L(\theta; \mathbf{x}) = \Big(\int c_1(\theta)\,d\theta\Big)\,T_n(\mathbf{x}) - \int \big(\theta + b(\theta)\big)c_1(\theta)\,d\theta + g(\mathbf{x}),$$

where $g(\mathbf{x})$ does not depend on θ. Exponentiating gives an exponential-family-type representation:

$$L(\theta; \mathbf{x}) = \exp\{\,Q(\theta)\,T_n(\mathbf{x}) - k(\theta) + g(\mathbf{x})\},$$

for suitable functions $Q(\theta) = \int c_1(\theta)d\theta$ and $k(\theta) = \int(\theta + b(\theta))c_1(\theta)\,d\theta$.

Remark.

$$\text{(uguaglianza di Cramér–Rao)} \iff \text{score affine in } T, \text{ più precisamente } U(\theta) \propto \big(T - \mathbb{E}_\theta[T]\big).$$

$$\text{Famiglia esponenziale (1 parametro)} \implies \text{score affine nel sufficiente } S_n.$$

Remark. In a one-parameter exponential family, the likelihood depends on the sample only through a finite-dimensional sufficient statistic (typically $S_n = \sum_{i=1}^n A(X_i)$). Hence, if a finite-sample efficient estimator exists, it can be taken as a function of that sufficient statistic. The connection between sufficiency and (unbiased) efficiency is often formalized via Rao–Blackwell and Lehmann–Scheffé.

5. **Newton–Raphson update hint for solving likelihood equations.**
The MLE $\hat{\theta}$ solves the likelihood equation

$$V_n(\theta) = \frac{\partial}{\partial\theta} \log L(\theta; \mathbf{x}) = 0,$$

with $\ell''(\hat{\theta}) < 0$ at a local maximum. A Newton–Raphson iteration is

$$\theta^{(m+1)} = \theta^{(m)} - \frac{\ell'(\theta^{(m)})}{\ell''(\theta^{(m)})} = \theta^{(m)} - \frac{V_n(\theta^{(m)})}{\ell''(\theta^{(m)})}.$$

A common scoring/Fisher-scoring variant replaces the observed second derivative by its expectation, $-I_n(\theta)$, giving

$$\boxed{\theta^{(m+1)} = \theta^{(m)} + \frac{V_n(\theta^{(m)})}{I_n(\theta^{(m)})}}$$

(under regularity, $I_n(\theta) = \mathbb{E}_\theta[-\ell''(\theta)]$).

L. Rao–Blackwell and Asymptotic Concepts

1. **Rao–Blackwell theorem (statement and role of sufficiency).** Let $(X_1, \ldots, X_n)$ be a random sample from a model F_θ, $\theta \in \Theta$ (regular or not). Let T_{1n} be a sufficient statistic for θ, and let T_{2n} be an unbiased estimator for θ (i.e. $\mathbb{E}[T_{2n}] = \theta$). Define the Rao–Blackwellized estimator

$$T_n := \mathbb{E}[T_{2n} \mid T_{1n}]\,.$$

Then:
(i) T_n is an estimator (it does not depend on θ);
(ii) T_n is unbiased: $\mathbb{E}_\theta[T_n] = \theta$;
(iii) $\mathrm{Var}_\theta(T_n) \le \mathrm{Var}_\theta(T_{2n})$.
Proof (as in the notes).
(i) **Unbiasedness:**

$$\mathbb{E}_\theta[T_n] = \mathbb{E}_\theta\big[\mathbb{E}_\theta\big(T_{2n} \mid T_{1n}\big)\big] = \mathbb{E}_\theta[T_{2n}] = \theta.$$

(ii) **Variance reduction:**

$$\mathrm{Var}_\theta(T_{2n}) = \mathrm{Var}_\theta\big(\mathbb{E}_\theta[T_{2n} \mid T_{1n}]\big) + \mathbb{E}_\theta\big[\mathrm{Var}_\theta(T_{2n} \mid T_{1n})\big] = \mathrm{Var}_\theta(T_n) + \mathbb{E}_\theta\big[\mathrm{Var}_\theta(T_{2n} \mid T_{1n})\big] \ge \mathrm{Var}_\theta(T_n).$$

(iii) T_n **is an estimator (does not depend on θ):** since T_{1n} is sufficient, the conditional distribution of $\mathbf{X}$ given T_{1n} does not depend on θ. Hence $\mathbb{E}_\theta(T_{2n} \mid T_{1n})$ is a function of T_{1n} only, i.e. it does not depend on θ.

2. **Variance decomposition used in the proof.** For any random variables Y and S,

$$\mathrm{Var}(Y) = \mathrm{Var}\big(\mathbb{E}[Y \mid S]\big) + \mathbb{E}\big[\mathrm{Var}(Y \mid S)\big].$$

3. **Asymptotic notions (as in the notes).**
(a) **Asymptotically correct.** An estimator T_n is asymptotically correct for θ if

$$\lim_{n\to\infty} \mathbb{E}[T_n] = \theta.$$

Example (from the notes). For the uncorrected sample variance $\tilde{S}_n^2$ in the Gaussian model,

$$\mathbb{E}[\tilde{S}_n^2] = \sigma^2 \frac{n-1}{n} \quad \Rightarrow \quad \lim_{n\to\infty} \mathbb{E}[\tilde{S}_n^2] = \sigma^2.$$

(b) **Consistency in mean square ⇒ consistency in probability.**
Mean-square consistency:

$$T_n \text{ is consistent in mean square if } \quad \lim_{n\to\infty} \mathbb{E}[(T_n-\theta)^2] = 0 \quad \text{(i.e. MSE}(T_n) \to 0).$$

Since $\text{MSE}(T_n) = \text{Var}(T_n) + b_n(\theta)^2$, MSE$\to 0$ implies $b_n(\theta) \to 0$, hence T_n is also asymptotically correct.
Consistency in probability:

$$T_n \xrightarrow{\mathbb{P}} \theta \quad \Longleftrightarrow \quad \lim_{n\to\infty} \mathbb{P}\big(|T_n-\theta| < \varepsilon\big) = 1, \quad \forall \varepsilon > 0.$$

Implication (Chebyshev, as in the notes): for any $\varepsilon > 0$,

$$\mathbb{P}\big(|T_n-\theta| \geq \varepsilon\big) \leq \frac{\mathbb{E}[(T_n-\theta)^2]}{\varepsilon^2}.$$

If $\mathbb{E}[(T_n-\theta)^2] \to 0$, then $\mathbb{P}(|T_n-\theta| \geq \varepsilon) \to 0$, i.e. $T_n \xrightarrow{\mathbb{P}} \theta$.

(c) **Strong (a.s.) consistency.** T_n is *strongly consistent* for θ_0 if

$$T_n \xrightarrow{a.s.} \theta_0 \quad \Longleftrightarrow \quad \mathbb{P}_{\theta_0}\Big(\lim_{n\to\infty} T_n = \theta_0\Big) = 1.$$

(d) **Asymptotic efficiency for unbiased estimators.** If T_n is unbiased for θ, then it is *asymptotically efficient* if

$$\lim_{n\to\infty} \text{Var}(T_n) = \frac{1}{I_n(\theta)}.$$

(e) **Asymptotic normality / asymptotic Gaussian standardization.** T_n is asymptotically Gaussian if

$$\frac{T_n - \mathbb{E}[T_n]}{\sqrt{\text{Var}(T_n)}} \xrightarrow{d} Z, \qquad Z \sim N(0,1).$$

M. Exercises and the Bayesian Approach

1. **Bernoulli model: MLE and properties.** Let $X_1, \ldots, X_n \overset{iid}{\sim} \text{Ber}(\theta)$ with $\theta \in (0,1)$ and observed $(x_1, \ldots, x_n)$.

   a. **MLE.**

$$\mathcal{L}(\theta; \underline{x}) = \prod_{i=1}^{n} \theta^{x_i}(1-\theta)^{1-x_i} = \theta^{\sum_{i=1}^n x_i}(1-\theta)^{n-\sum_{i=1}^n x_i}.$$

$$\ell(\theta) = \log \mathcal{L}(\theta; \underline{x}) = \Big(\sum_{i=1}^{n} x_i\Big)\log\theta + \Big(n - \sum_{i=1}^{n} x_i\Big)\log(1-\theta).$$

   Score and likelihood equation:

$$\ell'(\theta) = \frac{\sum x_i}{\theta} - \frac{n - \sum x_i}{1-\theta} = 0 \quad \Longrightarrow \quad \widehat{\theta}_{\text{ML}} = \frac{1}{n}\sum_{i=1}^{n} x_i = \bar{X}_n.$$

   Second derivative:

$$\ell''(\theta) = -\frac{\sum x_i}{\theta^2} - \frac{n - \sum x_i}{(1-\theta)^2} < 0,$$

   so this critical point is a maximum.

b. **Unbiasedness and variance.** Since $\mathbb{E}[X_i] = \theta$,

$$\mathbb{E}[\widehat{\theta}_{\mathrm{ML}}] = \mathbb{E}[\bar{X}_n] = \theta \quad \text{(unbiased)}.$$

Since $\mathrm{Var}(X_i) = \theta(1-\theta)$ and independence,

$$\mathrm{Var}(\widehat{\theta}_{\mathrm{ML}}) = \mathrm{Var}(\bar{X}_n) = \frac{\theta(1-\theta)}{n}.$$

c. **Fisher information.** For Bernoulli, the (sample) Fisher information is

$$I_n(\theta) = \mathrm{Var}(\ell'(\theta)) = -\mathbb{E}[\ell''(\theta)].$$

Using the expression for $\ell''(\theta)$ and $\mathbb{E}[\sum X_i] = n\theta$:

$$I_n(\theta) = -\mathbb{E}\left[-\frac{\sum X_i}{\theta^2} - \frac{n - \sum X_i}{(1-\theta)^2}\right] = \frac{n\theta}{\theta^2} + \frac{n(1-\theta)}{(1-\theta)^2} = \frac{n}{\theta(1-\theta)}.$$

d. **Efficiency (attains CRLB).** By the Cramér–Rao lower bound, for any unbiased estimator T_n of θ:

$$\mathrm{Var}(T_n) \geq \frac{1}{I_n(\theta)} = \frac{\theta(1-\theta)}{n}.$$

Since $\mathrm{Var}(\widehat{\theta}_{\mathrm{ML}}) = \theta(1-\theta)/n$, the MLE attains the CRLB and is efficient.

e. **Consistency via LLN.** By the (weak) law of large numbers,

$$\widehat{\theta}_{\mathrm{ML}} = \bar{X}_n \xrightarrow{\mathbb{P}} \mathbb{E}[X_1] = \theta.$$

f. **Asymptotic normality via CLT.** By the CLT,

$$\sqrt{n}(\widehat{\theta}_{\mathrm{ML}} - \theta) = \frac{\sum_{i=1}^n X_i - n\theta}{\sqrt{n}} \xrightarrow{d} \mathcal{N}\big(0, \theta(1-\theta)\big) = \mathcal{N}\Big(0, \frac{1}{I_1(\theta)}\Big).$$

2. **Beta–Bernoulli conjugacy and Bayes estimator.** Model: $X_1, \ldots, X_n \mid \theta \overset{iid}{\sim} \mathrm{Ber}(\theta)$, prior $\theta \sim \mathrm{Beta}(a, b)$ with $a, b > 0$.

Bayes' theorem (brief remark). In the continuous case,

$$\pi(\theta \mid x) = \frac{p(x \mid \theta)\,\pi(\theta)}{p(x)} \qquad \text{with} \qquad p(x) = \int p(x \mid t)\,\pi(t)\,dt,$$

where $p(x)$ is the marginal likelihood (evidence).

a. **Posterior update (including the missing normalization steps).** Let $S = \sum_{i=1}^n x_i$. The likelihood is

$$p(x_1, \ldots, x_n \mid \theta) = \prod_{i=1}^n \theta^{x_i}(1-\theta)^{1-x_i} = \theta^S(1-\theta)^{n-S}.$$

The $\mathrm{Beta}(a, b)$ prior density is

$$\pi(\theta) = \frac{1}{B(a,b)}\,\theta^{a-1}(1-\theta)^{b-1} = \frac{\Gamma(a+b)}{\Gamma(a)\Gamma(b)}\,\theta^{a-1}(1-\theta)^{b-1}, \qquad 0 < \theta < 1,$$

where $B(a, b) = \dfrac{\Gamma(a)\Gamma(b)}{\Gamma(a+b)}$.

By Bayes' theorem,

$$\pi(\theta \mid x_1, \dots, x_n) = \frac{p(x_1, \dots, x_n \mid \theta)\, \pi(\theta)}{\int_0^1 p(x_1, \dots, x_n \mid t)\, \pi(t)\, dt}.$$

The numerator is proportional to

$$\theta^S (1-\theta)^{n-S} \cdot \theta^{a-1}(1-\theta)^{b-1} = \theta^{a+S-1}(1-\theta)^{b+n-S-1}.$$

For the denominator,

$$\int_0^1 t^{a+S-1}(1-t)^{b+n-S-1}\, dt = B(a+S,\ b+n-S) = \frac{\Gamma(a+S)\Gamma(b+n-S)}{\Gamma(a+b+n)}.$$

Hence the normalized posterior density is

$$\pi(\theta \mid x_1, \dots, x_n) = \frac{\Gamma(a+b+n)}{\Gamma(a+S)\Gamma(b+n-S)}\, \theta^{a+S-1}(1-\theta)^{b+n-S-1},$$

so

$$\theta \mid x_1, \dots, x_n \sim \text{Beta}(a+S,\ b+n-S).$$

b. **Bayes estimator under squared loss.** Under squared loss, for a generic estimator $\widetilde{\theta}$ the posterior risk is

$$h(\widetilde{\theta}) = \mathbb{E}\big[(\widetilde{\theta} - \theta)^2 \mid x_1, \dots, x_n\big] = \int_0^1 (\widetilde{\theta} - \theta)^2\, \pi(\theta \mid x_1, \dots, x_n)\, d\theta.$$

Differentiating and setting to zero,

$$\frac{d}{d\widetilde{\theta}} h(\widetilde{\theta}) = 2\Big(\widetilde{\theta} - \mathbb{E}[\theta \mid x_1, \dots, x_n]\Big) = 0 \quad \Longrightarrow \quad \widetilde{\theta}^{\text{Bayes}} = \mathbb{E}[\theta \mid x_1, \dots, x_n].$$

Therefore,

$$\widetilde{\theta}_n^{\text{Bayes}} = \mathbb{E}[\theta \mid x_1, \dots, x_n] = \frac{a+S}{a+b+n}.$$

3. **Continuous model: $f(x;\theta) = 2\theta x e^{-\theta x^2}\mathbf{1}_{\mathbb{R}^+}(x)$.** Let $X_1, \dots, X_n$ be i.i.d. with density

$$f(x;\theta) = 2\theta x e^{-\theta x^2}\mathbf{1}_{(0,\infty)}(x), \qquad \theta > 0.$$

Denote $S_2 = \sum_{i=1}^n x_i^2$ and $S_1 = \sum_{i=1}^n x_i$.

a. **MLE.**

$$\mathcal{L}(\theta; \underline{x}) = \prod_{i=1}^n 2\theta x_i e^{-\theta x_i^2} = (2\theta)^n \Big(\prod_{i=1}^n x_i\Big) e^{-\theta S_2}.$$

Up to constants in θ,

$$\ell(\theta) = \log \mathcal{L}(\theta; \underline{x}) = n\log(2\theta) + \sum_{i=1}^n \log x_i - \theta S_2 = n\log\theta - \theta S_2 + \text{const}.$$

Score equation:

$$\ell'(\theta) = \frac{n}{\theta} - S_2 = 0 \quad \Longrightarrow \quad \widehat{\theta}_{\text{ML}} = \frac{n}{S_2}.$$

Since $\ell''(\theta) = -n/\theta^2 < 0$, this is a maximum.

b. **Moment-based estimator via $\mathbb{E}[X]$.** Compute $\mathbb{E}[X]$:

$$\mathbb{E}[X] = \int_0^\infty x\, 2\theta x e^{-\theta x^2}\, dx = \int_0^\infty 2\theta x^2 e^{-\theta x^2}\, dx.$$

With $y = x^2$ ($dy = 2x\, dx$), this yields (Gamma calculus as in the notes):

$$\mathbb{E}[X] = \frac{\sqrt{\pi}}{2\sqrt{\theta}}.$$

Equating $\mathbb{E}[X]$ to the sample mean $\bar{X}_n = S_1/n$ gives

$$\frac{\sqrt{\pi}}{2\sqrt{\theta}} = \frac{S_1}{n} \quad \Longrightarrow \quad \widetilde{\theta}_n = \frac{n^2\pi}{4S_1^2}.$$

c. **Sufficient statistic.** From the likelihood factorization,

$$\mathcal{L}(\theta; \underline{x}) = (2\theta)^n e^{-\theta S_2} \cdot \Big(\prod_{i=1}^n x_i\Big),$$

all dependence on θ is through $S_2 = \sum_{i=1}^n x_i^2$, hence S_2 is sufficient for θ (Fisher–Neyman factorization).

d. **Bayes estimator with exponential prior (squared loss) — all steps.** Assume the prior $\theta \sim \text{Exp}(\lambda)$ with density

$$\pi(\theta) = \lambda e^{-\lambda\theta} \mathbf{1}_{\{\theta>0\}}, \qquad \lambda > 0.$$

Step 1: Write Bayes' formula (continuous case).

$$\pi(\theta \mid \underline{x}) = \frac{\mathcal{L}(\theta; \underline{x})\, \pi(\theta)}{\int_0^\infty \mathcal{L}(t; \underline{x})\, \pi(t)\, dt}.$$

Step 2: Multiply likelihood and prior (kernel of the posterior). Using

$$\mathcal{L}(\theta; \underline{x}) = (2\theta)^n \Big(\prod_{i=1}^n x_i\Big) e^{-\theta S_2} \mathbf{1}_{\{\theta>0\}},$$

we get

$$\begin{aligned}\mathcal{L}(\theta; \underline{x})\, \pi(\theta) &= (2\theta)^n \Big(\prod_{i=1}^n x_i\Big) e^{-\theta S_2} \cdot \lambda e^{-\lambda\theta} \mathbf{1}_{\{\theta>0\}} \\ &= \lambda\, 2^n \Big(\prod_{i=1}^n x_i\Big)\, \theta^n\, e^{-(S_2+\lambda)\theta} \mathbf{1}_{\{\theta>0\}}.\end{aligned}$$

Therefore, up to a multiplicative constant not depending on θ,

$$\pi(\theta \mid \underline{x}) \propto \theta^n e^{-(S_2+\lambda)\theta} \mathbf{1}_{\{\theta>0\}}.$$

Step 3: Identify the posterior as a Gamma distribution (with normalization). Recall the Gamma(α, β) density (shape α, rate β):

$$g(\theta; \alpha, \beta) = \frac{\beta^\alpha}{\Gamma(\alpha)}\, \theta^{\alpha-1} e^{-\beta\theta} \mathbf{1}_{\{\theta>0\}}.$$

Match terms:

$$\theta^n = \theta^{(n+1)-1} \quad \Rightarrow \quad \alpha = n+1, \qquad e^{-(S_2+\lambda)\theta} \quad \Rightarrow \quad \beta = S_2 + \lambda.$$

Hence

$$\theta \mid \underline{x} \sim \text{Gamma}(n+1,\ S_2+\lambda),$$

and explicitly

$$\pi(\theta \mid \underline{x}) = \frac{(S_2+\lambda)^{n+1}}{\Gamma(n+1)}\,\theta^n e^{-(S_2+\lambda)\theta}\mathbf{1}_{\{\theta>0\}}.$$

Step 4: Bayes estimator under squared loss is the posterior mean. For squared loss $L(\widetilde{\theta},\theta) = (\widetilde{\theta}-\theta)^2$, the posterior risk is

$$h(\widetilde{\theta}) = \mathbb{E}\big[(\widetilde{\theta}-\theta)^2 \mid \underline{x}\big] = \int_0^\infty (\widetilde{\theta}-\theta)^2\,\pi(\theta \mid \underline{x})\,d\theta.$$

Differentiate and set to zero:

$$\frac{d}{d\widetilde{\theta}}h(\widetilde{\theta}) = 2\left(\widetilde{\theta} - \mathbb{E}[\theta \mid \underline{x}]\right) = 0 \quad\Longrightarrow\quad \widetilde{\theta}^{\text{Bayes}} = \mathbb{E}[\theta \mid \underline{x}].$$

Step 5: Compute the posterior mean. If $\Theta \sim \text{Gamma}(\alpha,\beta)$ (shape–rate), then

$$\mathbb{E}[\Theta] = \frac{\alpha}{\beta}.$$

With $\alpha = n+1$ and $\beta = S_2+\lambda$,

$$\widetilde{\theta}_n^{\text{Bayes}} = \mathbb{E}[\theta \mid \underline{x}] = \frac{n+1}{S_2+\lambda}.$$

4. **Uniform model: $U(a-b,a+b)$.** Let $X_1,\ldots,X_n \overset{iid}{\sim} U(a-b,a+b)$ with $a>0$, $b>0$. Let $X_{(1)} = \min_i X_i$ and $X_{(n)} = \max_i X_i$.

   a. **Likelihood and support constraints.** The density is $f(x;a,b) = \frac{1}{2b}\mathbf{1}_{(a-b,a+b)}(x)$, so

   $$\mathcal{L}(a,b;\underline{x}) = \left(\frac{1}{2b}\right)^n \prod_{i=1}^n \mathbf{1}_{(a-b,a+b)}(x_i).$$

   Therefore $\mathcal{L}(a,b;\underline{x}) > 0$ iff all points lie in the interval, i.e.

   $$X_{(1)} \geq a-b \quad\text{and}\quad X_{(n)} \leq a+b.$$

   Equivalently,

   $$a \leq X_{(1)}+b \quad\text{and}\quad a \geq X_{(n)}-b.$$

   b. **Sufficiency of $(X_{(1)},X_{(n)})$.** The likelihood depends on the sample only through $(X_{(1)},X_{(n)})$ via the indicator constraints, hence by factorization $(X_{(1)},X_{(n)})$ is sufficient for (a,b).

   c. **Moment-based estimators (as in the notes) and comparison.** Using

   $$\mathbb{E}[X] = a, \qquad \mathbb{E}[X^2] = a^2 + \frac{b^2}{3},$$

   the method of moments gives

   $$\widetilde{a}_n = \bar{X}_n = \frac{1}{n}\sum_{i=1}^n x_i, \qquad \widetilde{b}_n = \sqrt{3\left(\frac{1}{n}\sum_{i=1}^n x_i^2 - \bar{X}_n^{\,2}\right)}.$$

   These estimators use the full sample and are not (in general) functions of the sufficient statistic $(X_{(1)},X_{(n)})$.

d. **MLE via order statistics.** For fixed b, any a in the feasible interval

$$a \in [\, X_{(n)} - b,\ X_{(1)} + b \,]$$

gives the same likelihood value $(2b)^{-n}$ (since a appears only through feasibility). Thus maximizing $\mathcal{L}(a, b)$ reduces to minimizing b subject to feasibility:

$$X_{(n)} - b \le X_{(1)} + b \quad \Longleftrightarrow \quad b \ge \frac{X_{(n)} - X_{(1)}}{2}.$$

Hence

$$\widehat{b}_{\mathrm{ML}} = \frac{X_{(n)} - X_{(1)}}{2}.$$

With this choice, the feasible interval for a collapses to a single point:

$$\widehat{a}_{\mathrm{ML}} = \frac{X_{(1)} + X_{(n)}}{2}.$$

So the MLE is a function of $(X_{(1)}, X_{(n)})$.

N. MLE Properties

1. **Basic properties of the MLE (as referenced in the notes).**
   (a) **Sufficiency (regular models / exponential family settings).**
   In a (natural) exponential family, the likelihood depends on the sample through

$$\sum_{i=1}^{n} A(X_i),$$

   which is sufficient. Hence the MLE is a function of this sufficient statistic.
   (b) **Invariance property (MLE).**
   Let $X_1, \ldots, X_n$ be i.i.d. with likelihood $L(\theta; \underline{x})$, $\theta \in \Theta$, and let

$$\widehat{\theta}_n \in \arg\max_{\theta \in \Theta} L(\theta; \underline{x}) \qquad (\text{equivalently, } \widehat{\theta}_n \in \arg\max_{\theta \in \Theta} \ell(\theta))$$

   be an MLE of θ. If the parameter of interest is $\eta = \tau(\theta)$ for some function $\tau : \Theta \to \mathcal{H}$, then the invariance principle states that an MLE of η is obtained by applying the same transformation to the MLE of θ, namely

$$\widehat{\eta}_n = \widehat{\tau(\theta)} = \tau(\widehat{\theta}_n).$$

   If the MLE of θ is not unique, any maximizer $\widehat{\theta}_n$ produces an MLE $\widehat{\eta}_n = \tau(\widehat{\theta}_n)$ of η.
   (c) **Consistency (typical behavior).**
   In many models (notably exponential families), the MLE is typically strongly consistent. In simple examples (Bernoulli, Gaussian mean), this follows from the LLN because the MLE is a function of a sample mean.
   (d) **Cramér theorem (asymptotic normality, 1D regular case).**
   If $\widehat{\theta}_n$ is weakly consistent and the model is regular, then

$$\sqrt{n}\,(\widehat{\theta}_n - \theta_0) \xrightarrow{d} \mathcal{N}\left(0, \frac{1}{I(\theta_0)}\right).$$

   Consequences (as stated): asymptotically Gaussian, asymptotically correct, asymptotically efficient.

2. **Invariance principle: worked examples (with missing steps).**

a. **Survival / binomial reparametrization.**

Assume $X_1, \dots, X_n \overset{\text{iid}}{\sim} \text{Exp}(\theta)$ with rate $\theta > 0$, i.e.

$$f_\theta(x) = \theta e^{-\theta x} \mathbf{1}_{(0,\infty)}(x).$$

Fix a time $T > 0$ and define the survival indicator

$$Y_i = \mathbf{1}_{\{X_i \geq T\}} \in \{0, 1\}.$$

Then

$$p := \mathbb{P}_\theta(Y_i = 1) = \mathbb{P}_\theta(X_i \geq T) = 1 - F_\theta(T) = e^{-\theta T} \in (0, 1),$$

so $Y_i \sim \text{Bern}(p)$ and the number of survivors

$$Z = \sum_{i=1}^{n} Y_i$$

satisfies $Z \sim \text{Bin}(n, p)$.

Likelihood and MLE for p. For an observed value $Z = z \in \{0, \dots, n\}$,

$$L(p; z) = \binom{n}{z} p^z (1-p)^{n-z}.$$

The log-likelihood (up to an additive constant $\log \binom{n}{z}$) is

$$\ell(p; z) = z \log p + (n - z) \log(1 - p).$$

Differentiate and set to zero:

$$\ell'(p; z) = \frac{z}{p} - \frac{n-z}{1-p} = 0 \quad \Longrightarrow \quad z(1-p) = (n-z)p \quad \Longrightarrow \quad \widehat{p}_n = \frac{z}{n} = \frac{Z}{n}.$$

Invariance to get the MLE for θ. Since $p = g(\theta) = e^{-\theta T}$ is one-to-one for $\theta > 0$, invert it:

$$\theta = g^{-1}(p) = -\frac{1}{T} \log p.$$

By invariance,

$$\widehat{\theta}_n = g^{-1}(\widehat{p}_n) = -\frac{1}{T} \log\Big(\frac{Z}{n}\Big).$$

Boundary cases: if $Z = 0$ then $\widehat{p}_n = 0$ and the likelihood is maximized at $p \downarrow 0$, i.e. $\widehat{\theta}_n = +\infty$; if $Z = n$ then $\widehat{p}_n = 1$ and $\widehat{\theta}_n = 0$ (boundary of the parameter space if one requires $\theta > 0$ strictly).

b. **Two-group misclassification model.**

Let $X_i \in \{0, 1\}$ be the *observed* label, and suppose there is an unobserved true class $Y_i \in \{0, 1\}$ with

$$\mathbb{P}(Y_i = 1) = \theta, \qquad \theta \in (0, 1).$$

Assume a known classification accuracy $p \in (0, 1)$:

$$\mathbb{P}(X_i = Y_i) = p, \qquad \mathbb{P}(X_i \neq Y_i) = 1 - p,$$

so in particular

$$\mathbb{P}(X_i = 1 \mid Y_i = 1) = p, \qquad \mathbb{P}(X_i = 1 \mid Y_i = 0) = 1 - p.$$

By the law of total probability,

$$q := \mathbb{P}_\theta(X_i = 1) = \mathbb{P}(X_i = 1 \mid Y_i = 1)\mathbb{P}(Y_i = 1) + \mathbb{P}(X_i = 1 \mid Y_i = 0)\mathbb{P}(Y_i = 0) = \theta p + (1-p)(1-\theta).$$

Rewrite q as an affine function of θ:

$$q = \theta p + (1-p)(1-\theta) = \theta(p-(1-p)) + (1-p) = (2p-1)\theta + (1-p).$$

If $p = \frac{1}{2}$ then $2p-1 = 0$ and $q = 1-p = \frac{1}{2}$ does not depend on θ (non-identifiable), so assume $p \neq \frac{1}{2}$.

MLE for q (Bernoulli model). Since $X_i \overset{\text{iid}}{\sim} \text{Bern}(q)$, for $S = \sum_{i=1}^n X_i$ the likelihood is

$$L(q; S) = q^S (1-q)^{n-S}, \qquad q \in (0,1),$$

hence the MLE is

$$\widehat{q}_n = \bar{X}_n = \frac{1}{n}\sum_{i=1}^n X_i.$$

Invariance to get the MLE for θ. Solve $q = (2p-1)\theta + (1-p)$ for θ:

$$\theta = \frac{q-(1-p)}{2p-1}.$$

By invariance,

$$\widehat{\theta}_n = \frac{\widehat{q}_n - (1-p)}{2p-1}, \qquad p \neq \frac{1}{2}.$$

If you impose the constraint $\theta \in [0,1]$, the constrained MLE is the truncated version $\widehat{\theta}_n^c = \min\{1, \max\{0, \widehat{\theta}_n\}\}$.

3. **Definitions: weak consistency vs. strong consistency.**
Let $(X_1, \ldots, X_n)$ be a random sample from $X \sim f(x;\theta)$ with $\theta \in \Theta$. Let $T_n = T(X_1, \ldots, X_n)$ be an estimator for θ, and let θ_0 be the true parameter value.
   a. **Weak consistency.**
   T_n is weakly consistent for θ if
   $$T_n \xrightarrow{\mathbb{P}} \theta_0 \quad (n \to \infty),$$
   equivalently,
   $$\lim_{n\to\infty} \mathbb{P}_{\theta_0}\big(|T_n - \theta_0| > \varepsilon\big) = 0 \quad \text{for any } \varepsilon > 0 \text{ and any } \theta_0 \in \Theta.$$
   b. **Strong consistency.**
   T_n is strongly consistent for θ_0 if
   $$T_n \xrightarrow{a.s.} \theta_0 \quad (n \to \infty),$$
   equivalently,
   $$\mathbb{P}_{\theta_0}\Big(\lim_{n\to\infty} T_n = \theta_0\Big) = 1,$$
   or, for every $\varepsilon > 0$,
   $$\lim_{n\to\infty} \mathbb{P}_{\theta_0}\Big(\forall m \geq n : \ |T_m - \theta_0| < \varepsilon\Big) = 1.$$

4. **Remark: MLEs typically strongly consistent (as stated) + basic examples.**
Remark. In general, MLEs are strongly consistent.
   a. $(X_1, \ldots, X_n)$ from $X \sim \mathcal{N}(\theta, 1)$.
   $$\widehat{\theta}_n = \frac{1}{n}\sum_{i=1}^n X_i \xrightarrow{a.s.} \theta = \mathbb{E}[X].$$
   b. $(X_1, \ldots, X_n)$ from $X \sim \text{Ber}(\theta)$.
   $$\widehat{\theta}_n = \frac{1}{n}\sum_{i=1}^n X_i \xrightarrow{a.s.} \theta = \mathbb{E}[X].$$

c. **Exponential family.**
The MLE is a function of $\frac{1}{n}\sum_{i=1}^n A(X_i)$, and

$$\frac{1}{n}\sum_{i=1}^n A(X_i) \xrightarrow{a.s.} \mathbb{E}[A(X)] \qquad (n \to \infty).$$

Also (as referenced in the notes):

$$\frac{1}{n}\sum_{i=1}^n X_i \to \mathbb{E}[X] = \theta \quad a.s., \qquad \sup_{\theta\in\Theta}\left|\frac{1}{n}\sum_{i=1}^n X_i - \theta\right| \to 0 \quad \text{as } n \to \infty$$

(*uniform law of large numbers*).

5. **Strong consistency sketch (log-likelihood approach).**
Let $(X_1, \ldots, X_n)$ be a random sample from a model with densities $f(\,\cdot\,;\theta)$, $\theta \in \Theta$, and let

$$\ell_n(\theta) := \sum_{i=1}^n \log f(X_i;\theta)$$

be the log-likelihood. Fix the true parameter $\theta_0 \in \Theta$.

a. **Normalized log-likelihood difference.**
For any $\theta \in \Theta$ consider the normalized difference

$$\frac{1}{n}\big(\ell_n(\theta) - \ell_n(\theta_0)\big) = \frac{1}{n}\sum_{i=1}^n \log\frac{f(X_i;\theta)}{f(X_i;\theta_0)}.$$

b. **Pointwise strong LLN limit.**
Assume $\mathbb{E}_{\theta_0}\left[\left|\log\frac{f(X;\theta)}{f(X;\theta_0)}\right|\right] < \infty$. Then for each fixed θ the strong LLN gives

$$\frac{1}{n}\sum_{i=1}^n \log\frac{f(X_i;\theta)}{f(X_i;\theta_0)} \xrightarrow{a.s.} \mathbb{E}_{\theta_0}\left[\log\frac{f(X;\theta)}{f(X;\theta_0)}\right].$$

c. **Strict negativity for $\theta \neq \theta_0$ (Jensen / identifiability).**
If $\theta \neq \theta_0$ and the model is identifiable, then by Jensen's inequality

$$\mathbb{E}_{\theta_0}\left[\log\frac{f(X;\theta)}{f(X;\theta_0)}\right] \le \log \mathbb{E}_{\theta_0}\left[\frac{f(X;\theta)}{f(X;\theta_0)}\right].$$

Reminder: what does $\mathbb{E}_{\theta_0}[\cdot]$ mean? Under $\mathbb{P}_{\theta_0}$ the r.v. X has density $f(\,\cdot\,;\theta_0)$, hence for any measurable g,

$$\mathbb{E}_{\theta_0}[g(X)] = \int g(x)\, f(x;\theta_0)\, dx.$$

Applying this with $g(x) = \dfrac{f(x;\theta)}{f(x;\theta_0)}$ gives

$$\mathbb{E}_{\theta_0}\left[\frac{f(X;\theta)}{f(X;\theta_0)}\right] = \int \frac{f(x;\theta)}{f(x;\theta_0)}\, f(x;\theta_0)\, dx = \int f(x;\theta)\, dx = 1,$$

so

$$\mathbb{E}_{\theta_0}\left[\log\frac{f(X;\theta)}{f(X;\theta_0)}\right] \le \log(1) = 0,$$

and the inequality is strict for $\theta \neq \theta_0$. Hence the limit is < 0 and therefore, for each fixed $\theta \neq \theta_0$,

$$\ell_n(\theta) - \ell_n(\theta_0) \xrightarrow{a.s.} -\infty \qquad \text{equivalently} \qquad \ell_n(\theta_0) - \ell_n(\theta) \xrightarrow{a.s.} +\infty.$$

d. **From pointwise separation to $\widehat{\theta}_n \to \theta_0$.**
The previous step gives *pointwise* separation: for each fixed $\theta \neq \theta_0$, eventually $\ell_n(\theta_0) > \ell_n(\theta)$. To conclude that any MLE $\widehat{\theta}_n \in \arg\max_{\theta\in\Theta} \ell_n(\theta)$ satisfies $\widehat{\theta}_n \to \theta_0$, one needs this separation to hold *simultaneously* over all competitors.
- If Θ is *finite*, simultaneity follows by intersecting finitely many almost-sure events (equivalently, one may write an explicit probabilistic bound via a union bound, see below).
- If Θ is *infinite*, one typically needs a *uniform* LLN (or another uniform convergence argument) to control

$$\sup_{\theta\in\Theta} \left| \frac{1}{n}\ell_n(\theta) - \mathbb{E}_{\theta_0}[\log f(X;\theta)] \right|$$

and then deduce strong consistency of the MLE.

6. **Finite parameter space: explicit union-bound argument (as in lecture).**
Assume Θ is finite:

$$\Theta = \{\theta_0, \theta_1, \ldots, \theta_m\}, \qquad m \in \mathbb{N}.$$

Fix $\varepsilon > 0$ and $\delta > 0$. For each competitor θ_j and each $n_0 \in \mathbb{N}$ define

$$A_{j,n_0} := \Big\{ \ell_n(\theta_0) - \ell_n(\theta_j) > \varepsilon,\ \forall n \geq n_0 \Big\}, \qquad j = 1, \ldots, m.$$

For each fixed j we have $\ell_n(\theta_0) - \ell_n(\theta_j) \xrightarrow{a.s.} +\infty$, hence

$$\mathbb{P}\Big(\bigcup_{n_0=1}^{\infty} A_{j,n_0} \Big) = 1 \quad \Longrightarrow \quad \exists\, n_0(j) \text{ s.t. } \mathbb{P}(A_{j,n_0(j)}) > 1 - \frac{\delta}{m}.$$

Now set the common threshold $n_0 := \max_{1\leq j\leq m} n_0(j)$. Since $A_{j,n_0} \supseteq A_{j,n_0(j)}$,

$$\mathbb{P}(A_{j,n_0}) \geq \mathbb{P}(A_{j,n_0(j)}) > 1 - \frac{\delta}{m}, \qquad j = 1, \ldots, m.$$

By the union bound,

$$\mathbb{P}\Big(\bigcap_{j=1}^{m} A_{j,n_0} \Big) \geq 1 - \sum_{j=1}^{m} \mathbb{P}(A_{j,n_0}^c) > 1 - m \cdot \frac{\delta}{m} = 1 - \delta.$$

On $\bigcap_{j=1}^m A_{j,n_0}$ we have, for all $n \geq n_0$ and all $j \neq 0$, $\ell_n(\theta_0) > \ell_n(\theta_j)$, hence $\widehat{\theta}_n = \theta_0$ for all large n. Therefore $\widehat{\theta}_n \xrightarrow{a.s.} \theta_0$ when Θ is finite.

O. o_p, O_p, Taylor in Probability, Delta Method, Cramér Theorem

1. **Definitions of $o_p(\tau_n)$ and $O_p(\tau_n)$; special case $\tau_n = 1$.**
Let $(X_n)_{n\geq 1}$ be a sequence of random variables and $(\tau_n)_{n\geq 1}$ a sequence of positive real numbers.

(i) $X_n = o_p(\tau_n)$ (smaller order than τ_n in probability) if

$$\frac{X_n}{\tau_n} \xrightarrow{\mathbb{P}} 0 \qquad (n \to \infty).$$

(ii) $X_n = O_p(\tau_n)$ (at most the same order as τ_n in probability) if

$$\forall \varepsilon > 0,\ \exists M_\varepsilon :\ \mathbb{P}\left(\left| \frac{X_n}{\tau_n} \right| > M_\varepsilon \right) < \varepsilon, \quad \forall n > M_\varepsilon,$$

where M_ε may depend on ε.

Remark (case $\tau_n = 1$).

$$X_n = o_p(1) \iff X_n \xrightarrow{\mathbb{P}} 0, \qquad X_n = O_p(1) \iff X_n \text{ does not explode to } +\infty \text{ in probability.}$$

2. **Algebra rules for o_p and O_p. (dim solo (i))**
Let (X_n), (Y_n) be sequences of random variables and let (τ_n), (s_n) be positive real sequences.

(i) If $X_n = o_p(\tau_n)$ and $Y_n = o_p(s_n)$, then

$$X_n Y_n = o_p(\tau_n s_n), \qquad X_n + Y_n = o_p\big(\max\{\tau_n, s_n\}\big).$$

Proof. Since $X_n = o_p(\tau_n)$ and $Y_n = o_p(s_n)$, we have $\frac{X_n}{\tau_n} \xrightarrow{p} 0$ and $\frac{Y_n}{s_n} \xrightarrow{p} 0$, i.e. for every $\varepsilon > 0$, $\mathbb{P}\Big(\Big|\frac{X_n}{\tau_n}\Big| > \varepsilon\Big) \to 0$ and $\mathbb{P}\Big(\Big|\frac{Y_n}{s_n}\Big| > \varepsilon\Big) \to 0$.

Product. For any $\varepsilon > 0$,

$$\mathbb{P}\left(\left|\frac{X_n Y_n}{\tau_n s_n}\right| > \varepsilon\right) = \mathbb{P}\left(\left|\frac{X_n}{\tau_n}\right|\left|\frac{Y_n}{s_n}\right| > \varepsilon\right) \le \mathbb{P}\left(\left|\frac{X_n}{\tau_n}\right| > \sqrt{\varepsilon}\right) + \mathbb{P}\left(\left|\frac{Y_n}{s_n}\right| > \sqrt{\varepsilon}\right) \to 0,$$

hence $X_n Y_n = o_p(\tau_n s_n)$.

Sum. Let $q_n := \max\{\tau_n, s_n\}$, so that $\tau_n \le q_n$ and $s_n \le q_n$. Then for any $\varepsilon > 0$,

$$\mathbb{P}\left(\left|\frac{X_n + Y_n}{q_n}\right| > \varepsilon\right) \le \mathbb{P}\left(\left|\frac{X_n}{q_n}\right| > \frac{\varepsilon}{2}\right) + \mathbb{P}\left(\left|\frac{Y_n}{q_n}\right| > \frac{\varepsilon}{2}\right) \le \mathbb{P}\left(\left|\frac{X_n}{\tau_n}\right| > \frac{\varepsilon}{2}\right) + \mathbb{P}\left(\left|\frac{Y_n}{s_n}\right| > \frac{\varepsilon}{2}\right) \to 0,$$

so $X_n + Y_n = o_p(q_n) = o_p(\max\{\tau_n, s_n\})$. □

(ii) If $X_n = O_p(\tau_n)$ and $Y_n = O_p(s_n)$, then

$$X_n Y_n = O_p(\tau_n s_n), \qquad X_n + Y_n = O_p\big(\max\{\tau_n, s_n\}\big).$$

(iii) If $X_n = O_p(\tau_n)$ and $Y_n = o_p(s_n)$, then

$$X_n Y_n = o_p(\tau_n s_n).$$

3. **Theorem: convergence in distribution implies an O_p bound.**
Theorem. Let $(X_n)_n$ be a sequence of r.v.'s and $(r_n)_n$ a sequence of real numbers such that

$$r_n(X_n - c) \xrightarrow{d} Z \qquad (n \to \infty),$$

where c is a constant and Z is a non-degenerate r.v. Then

$$X_n = c + O_p(r_n^{-1}).$$

Proof. Let F be the cdf of Z and let $t > 0$ be a continuity point of the distribution of $|Z|$. Set

$$Y_n := r_n(X_n - c).$$

The assumption $r_n(X_n - c) \xrightarrow{d} Z$ is exactly $Y_n \xrightarrow{d} Z$.

Define the tail function

$$G(t) := \mathbb{P}(|Z| > t).$$

Since Z is a (real-valued) random variable, $|Z| < \infty$ a.s., hence $G(t) \downarrow 0$ as $t \to \infty$.

By convergence in distribution, for every continuity point t of $|Z|$ we have

$$\mathbb{P}(|Y_n| > t) \longrightarrow \mathbb{P}(|Z| > t) = G(t).$$

Fix such a continuity point t. Define a nonnegative sequence $(s_n(t))_{n\geq 1}$ by

$$s_n(t) := \sup_{m\geq n} \left|\mathbb{P}(|Y_m| > t) - G(t)\right|.$$

Then $s_n(t) \downarrow 0$ as $n \to \infty$, and for all n we have

$$\mathbb{P}(|Y_n| > t) \leq G(t) + s_n(t).$$

Fix $\varepsilon > 0$. Choose $t_\varepsilon > 0$ such that $G(t_\varepsilon) < \varepsilon$, and (if necessary) increase t_ε slightly so that it is a continuity point of the distribution of $|Z|$. Consider the sequence $s_n := s_n(t_\varepsilon)$; then $s_n \to 0$, so there exists n_0 such that $s_n < \varepsilon - G(t_\varepsilon)$ for all $n \geq n_0$. Hence, for all $n \geq n_0$,

$$\mathbb{P}(|Y_n| > t_\varepsilon) \leq G(t_\varepsilon) + s_n < \varepsilon.$$

Thus, with $M := t_\varepsilon$,

$$\mathbb{P}\big(|r_n(X_n - c)| > M\big) < \varepsilon \qquad \text{for all } n \geq n_0,$$

which is exactly $r_n(X_n - c) = O_p(1)$. Therefore,

$$r_n(X_n - c) = O_p(1) \quad \Longleftrightarrow \quad X_n - c = O_p(r_n^{-1}),$$

and hence $X_n = c + O_p(r_n^{-1})$. □

4. **Taylor expansion in probability with remainder control.**
Theorem (Taylor in probability). Let $(X_n)_n$ be a sequence of r.v.s such that

$$X_n = c + O_p(r_n),$$

with c a constant and $(r_n)_n$ a sequence of real numbers such that $r_n \to 0$. If $g(\cdot)$ is a function with k continuous derivatives in a neighborhood of c, then

$$g(X_n) = g(c) + g'(c)(X_n - c) + \cdots + \frac{g^{(k)}(c)}{k!}(X_n - c)^k + o_p(r_n^k).$$

Proof. Apply Taylor's theorem with Lagrange remainder around c up to order k. For each n there exists a (possibly random) point Z_n lying between c and X_n such that

$$g(X_n) = g(c) + g'(c)(X_n - c) + \cdots + \frac{g^{(k)}(c)}{k!}(X_n - c)^k + R_n,$$

where

$$R_n = \frac{(X_n - c)^k}{k!}\big(g^{(k)}(Z_n) - g^{(k)}(c)\big).$$

It remains to show that $R_n = o_p(r_n^k)$. Since $X_n = c + O_p(r_n)$ and $r_n \to 0$, we have $X_n - c = o_p(1)$, hence $X_n \xrightarrow{p} c$. Moreover, because Z_n lies between c and X_n,

$$|Z_n - c| \leq |X_n - c|,$$

so $Z_n - c = o_p(1)$ and therefore $Z_n \xrightarrow{p} c$. By continuity of $g^{(k)}$ at c, the continuous mapping theorem gives

$$g^{(k)}(Z_n) \xrightarrow{p} g^{(k)}(c) \quad \Longrightarrow \quad g^{(k)}(Z_n) - g^{(k)}(c) = o_p(1).$$

Also, from $X_n - c = O_p(r_n)$ we obtain $(X_n - c)^k = O_p(r_n^k)$. Combining these two facts,

$$R_n = \frac{(X_n - c)^k}{k!}\big(g^{(k)}(Z_n) - g^{(k)}(c)\big) = O_p(r_n^k) \cdot o_p(1) = o_p(r_n^k).$$

This proves the claimed expansion. □

5. **Delta method and applications to transformations $g(\widehat{\theta}_n)$.**
Corollary (Delta method). Let $(X_n)_n$ be a sequence of random variables such that

$$\sqrt{n}\,(X_n - c) \xrightarrow{d} Z,$$

where $c \in \mathbb{R}$ is a constant and Z is a non-degenerate random variable. If g is differentiable at c and g' is continuous in a neighborhood of c (equivalently, $g \in C^1$ in a neighborhood of c), then

$$\sqrt{n}\,\big(g(X_n) - g(c)\big) \xrightarrow{d} g'(c)\,Z.$$

Application to MLE transforms (invariance + delta). If $\widehat{\theta}_n$ is the MLE of θ, then by invariance $g(\widehat{\theta}_n)$ is the MLE of $g(\theta)$, and if

$$\sqrt{n}(\widehat{\theta}_n - \theta_0) \xrightarrow{d} \mathcal{N}\left(0, \frac{1}{I(\theta_0)}\right),$$

then

$$\sqrt{n}\big(g(\widehat{\theta}_n) - g(\theta_0)\big) \xrightarrow{d} \mathcal{N}\left(0, \frac{(g'(\theta_0))^2}{I(\theta_0)}\right).$$

6. **Cramér theorem (MLE asymptotic normality) in 1D regular models.**

(a) **Statement + regularity conditions (as in notes).**
Consider a random sample $(X_1, \dots, X_n)$ from a regular model $X \sim f(x;\theta)$, $\theta \in \Theta \subset \mathbb{R}$. Let $\widehat{\theta}_n$ be the MLE. If $\widehat{\theta}_n$ is weakly consistent for θ_0, then

$$\sqrt{n}(\widehat{\theta}_n - \theta_0) \xrightarrow{d} \mathcal{N}\left(0, \frac{1}{I(\theta_0)}\right), \qquad (n \to \infty),$$

where $I(\theta_0)$ is the Fisher information of the model.

Regularity conditions (as written in the notes):

- derivatives of $f(x;\theta)$ w.r.t. θ exist up to order 3;
- there exist functions $g(\cdot), h(\cdot), H(\cdot)$ such that for all x and for θ in a neighborhood of θ_0,

$$\left|\frac{\partial}{\partial\theta} f(x;\theta)\right| \le g(x), \quad \left|\frac{\partial^2}{\partial\theta^2} f(x;\theta)\right| \le h(x),$$

and (as used in the remainder bound)

$$\left|\frac{\partial^3}{\partial\theta^3} \log f(x;\theta)\right| \le H(x), \qquad \mathbb{E}_{\theta_0}[H(X)] < M_0 < +\infty.$$

(b) **(a) Taylor expansion of the score around θ_0.**
Let $\ell(\theta) = \log \mathcal{L}(\theta; x)$ and $\ell'(\theta) = \frac{d}{d\theta}\ell(\theta)$. Taylor-expand $\ell'(\theta)$ around θ_0:

$$\ell'(\theta) = \ell'(\theta_0) + \ell''(\theta_0)(\theta - \theta_0) + \frac{1}{2}\,\ell'''(\widetilde{\theta})(\theta - \theta_0)^2, \qquad \widetilde{\theta} \in (\theta, \theta_0).$$

Replace θ with $\widehat{\theta}_n$ (score equation $\ell'(\widehat{\theta}_n) = 0$):

$$0 = \ell'(\widehat{\theta}_n) = \ell'(\theta_0) + (\widehat{\theta}_n - \theta_0)\left[\ell''(\theta_0) + \frac{1}{2}\,\ell'''(\widehat{\widetilde{\theta}})(\widehat{\theta}_n - \theta_0)\right],$$

with $\widehat{\widetilde{\theta}} \in (\widehat{\theta}_n, \theta_0)$. Hence

$$\widehat{\theta}_n - \theta_0 = -\frac{\ell'(\theta_0)}{\ell''(\theta_0) + \frac{1}{2}\,\ell'''(\widehat{\widetilde{\theta}})(\widehat{\theta}_n - \theta_0)}.$$

Multiply by $\sqrt{n}$:

$$\sqrt{n}(\widehat{\theta}_n - \theta_0) = \frac{-\frac{1}{\sqrt{n}}\ell'(\theta_0)\ (a)}{\frac{1}{n}\ell''(\theta_0)\ (b)\ +\ \frac{\ell'''(\widehat{\widetilde{\theta}})}{2n}(\widehat{\theta}_n - \theta_0)\ (c)}.$$

(c) **(b) CLT for the score; LLN for observed information; remainder $o_p(1)$.**

(a) CLT for the score.

$$-\frac{1}{\sqrt{n}}\ell'(\theta_0) = -\frac{1}{\sqrt{n}}\sum_{i=1}^{n}\underbrace{\frac{\frac{\partial}{\partial\theta}f(X_i;\theta_0)}{f(X_i;\theta_0)}}_{=:w_i}.$$

Notes (as written): $w_i \overset{d}{=} w_j$, $w_i \perp w_j$ for $i \neq j$, $\mathbb{E}[w_i] = 0$, $\mathrm{Var}(w_i) = I(\theta_0)$. By the CLT,

$$-\frac{1}{\sqrt{n}}\ell'(\theta_0) \xrightarrow{d} \mathcal{N}(0, I(\theta_0)).$$

(b) LLN for the observed information term.

$$\frac{1}{n}\ell''(\theta_0) = \frac{1}{n}\sum_{i=1}^{n}\underbrace{\frac{\frac{\partial^2}{\partial\theta^2}f(X_i;\theta_0)}{f(X_i;\theta_0)}}_{=:w_i^*} - \frac{1}{n}\sum_{i=1}^{n} w_i^2.$$

Notes (as written): w_i^*, w_i are i.i.d., $\mathbb{E}[w_i^*] = 0$. By the strong LLN,

$$\frac{1}{n}\sum_{i=1}^{n} w_i^* \xrightarrow{a.s.} 0, \qquad \frac{1}{n}\sum_{i=1}^{n} w_i^2 \xrightarrow{a.s.} I(\theta_0),$$

hence

$$\frac{1}{n}\ell''(\theta_0) \xrightarrow{a.s.} -I(\theta_0).$$

(c) Remainder is $o_p(1)$. Consider

$$\frac{\ell'''(\widehat{\widehat{\theta}})}{2n}(\widehat{\theta}_n - \theta_0), \qquad \widehat{\widehat{\theta}} \in (\widehat{\theta}_n, \theta_0).$$

Since $\widehat{\theta}_n \xrightarrow{p} \theta_0$, we have $(\widehat{\theta}_n - \theta_0) = o_p(1)$. Moreover, by the bound on $\partial_\theta^3 \log f$,

$$\frac{1}{2}\left|\ell'''(\widehat{\widehat{\theta}})\right| = \frac{1}{2}\left|\sum_{i=1}^{n}\frac{\partial^3}{\partial\theta^3}\log f(X_i;\widehat{\widehat{\theta}})\right| \leq \sum_{i=1}^{n} H(X_i),$$

hence

$$\frac{1}{2n}\left|\ell'''(\widehat{\widehat{\theta}})\right| \leq \frac{1}{n}\sum_{i=1}^{n} H(X_i) \xrightarrow{a.s.} \mathbb{E}_{\theta_0}[H(X)] < +\infty,$$

so $\frac{1}{2n}\ell'''(\widehat{\widehat{\theta}}) = O_p(1)$ and therefore

$$\frac{\ell'''(\widehat{\widehat{\theta}})}{2n}(\widehat{\theta}_n - \theta_0) = o_p(1).$$

Combining (b)+(c),

$$\frac{1}{n}\ell''(\theta_0) + \frac{\ell'''(\widehat{\widehat{\theta}})}{2n}(\widehat{\theta}_n - \theta_0) \xrightarrow{p} -I(\theta_0) \neq 0.$$

By Slutsky's theorem,

$$\sqrt{n}(\widehat{\theta}_n - \theta_0) \xrightarrow{d} \mathcal{N}\left(0, \frac{1}{I(\theta_0)}\right).$$

(d) **(c) Consequences: asymptotic normality, correctness, efficiency; invariance + delta method.**

Consequences (as listed in the notes):

- **MLE asymptotically Gaussian:** $\sqrt{n}(\widehat{\theta}_n - \theta_0) \Rightarrow \mathcal{N}(0, 1/I(\theta_0))$.

- **MLE asymptotically correct:** typically $\mathbb{E}[\widehat{\theta}_n] \to \theta_0$ (asymptotically unbiased/correct).

- **MLE asymptotically efficient:** $\mathrm{Var}(\widehat{\theta}_n) \sim \frac{1}{nI(\theta_0)}$, attaining the Cramér–Rao bound asymptotically.

Invariance property and Delta method (as in the notes). If we are interested in estimating $g(\theta)$, then, by the invariance property of the MLE, the MLE of $g(\theta)$ is $g(\widehat{\theta}_n)$. Moreover, by the Delta method we have a CLT for $g(\widehat{\theta}_n)$:

$$\sqrt{n}\big(g(\widehat{\theta}_n) - g(\theta_0)\big) \xrightarrow{d} g'(\theta_0)\, R, \qquad n \to \infty,$$

where

$$R \sim \mathcal{N}\left(0, \frac{1}{I(\theta_0)}\right).$$

P. Hypothesis Testing: Basics

1. **Testing: setting and goal.**
   We observe data points $(x_1, \ldots, x_n)$ and model $(X_1, \ldots, X_n)$ as i.i.d. random variables from F_θ with $\theta \in \Theta$. If we know θ, we know the population (distribution). The goal is learning θ. *Point estimation* assigns a value to θ, while *hypothesis testing* tests a particular value (or subset) for θ. A test is built from: (i) a statistical hypothesis, (ii) a random sample, (iii) a decision rule.
2. **Hypotheses: simple vs composite (one-sided and two-sided).**
   A *statistical hypothesis* is a statement that specifies a value or a collection of values for the parameter $\theta \in \Theta$.
   - **Simple hypothesis:** $\theta = \theta^* \in \Theta$.
   - **Composite one-sided (unidirectional):** $\theta \geq \theta^*$ or $\theta \leq \theta^*$.
   - **Composite two-sided (bidirectional):** $\theta \neq \theta^*$.

   We specify two hypotheses (one is the complement of the other) and partition the parameter space:

   $$H_0 :\ \theta \in \Theta_0 \subseteq \Theta, \qquad H_1 :\ \theta \notin \Theta_0\ (= \theta \in \Theta_0^c), \qquad \Theta_0 \cup \Theta_0^c = \Theta.$$

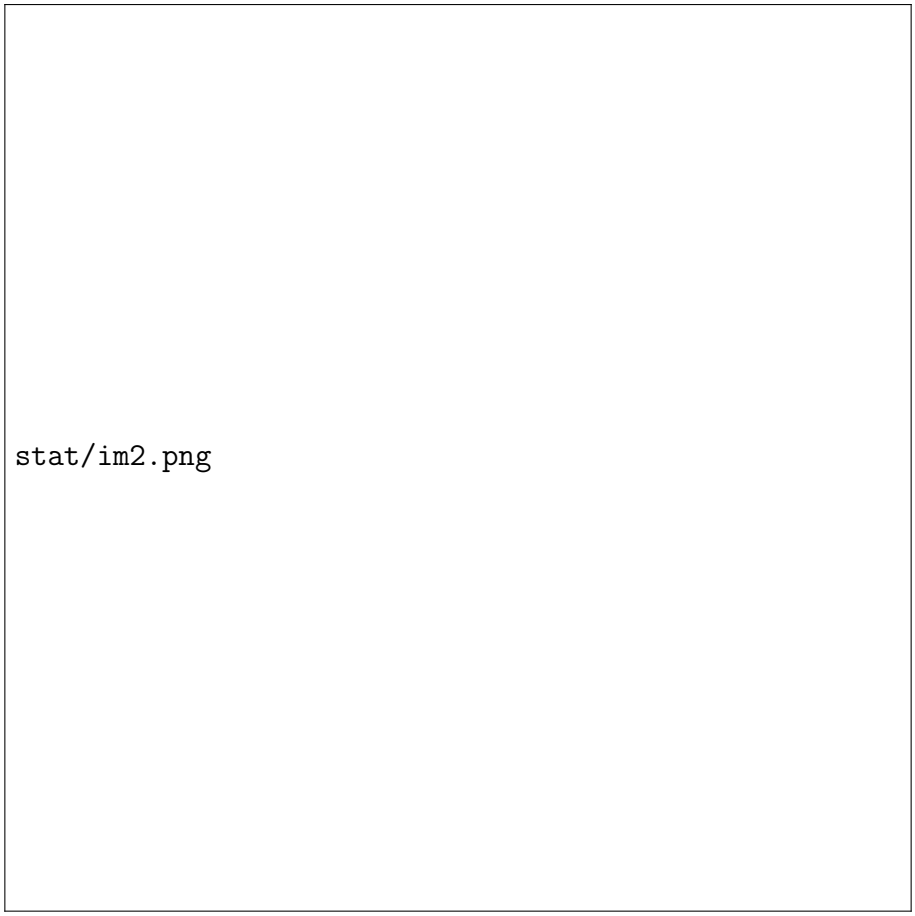
   Example (fair coin).

   $$H_0 :\ \theta = \frac{1}{2}, \qquad H_1 :\ \theta \neq \frac{1}{2}.$$

3. **Decision rule and rejection region R_0.**
   Let $(X_1, \ldots, X_n)$ be an i.i.d. random sample from F_θ ($\theta \in \Theta$), and let $T_n = T(X_1, \ldots, X_n)$ be a statistic. A *decision rule* is specified by a *rejection region* R_0:

   $$\text{Reject } H_0 \text{ if } T_n \in R_0, \qquad \text{Do not reject } H_0 \text{ if } T_n \notin R_0\ (= R_0^c).$$

   Thus, the statistic (or sample) space is partitioned into R_0 (reject) and R_0^c (do not reject).



4. **Type I error α, type II error β; power $1-\beta$.**
   There are four possible outcomes:
   - correct c_1: do not reject H_0 and H_0 is true;
   - correct c_2: reject H_0 and H_0 is false;
   - error e_1 (Type I): reject H_0 and H_0 is true;
   - error e_2 (Type II): do not reject H_0 and H_0 is false.

   For a fixed θ, consider the rejection region R_0 (reject H_0 if $X \in R_0$). Then:

$$\alpha(\theta) = \mathbb{P}_\theta(e_1) = \mathbb{P}_\theta(X \in R_0), \qquad \theta \in \Theta_0,$$
$$\beta(\theta) = \mathbb{P}_\theta(e_2) = \mathbb{P}_\theta(X \in R_0^c), \qquad \theta \in \Theta_0^c,$$
$$1-\alpha(\theta) = \mathbb{P}_\theta(c_1) = \mathbb{P}_\theta(X \in R_0^c), \qquad \theta \in \Theta_0,$$

   and, defining the **power** as $\delta(\theta) = 1-\beta(\theta)$,

$$\delta(\theta) = 1-\beta(\theta) = \mathbb{P}_\theta(c_2) = \mathbb{P}_\theta(X \in R_0), \qquad \theta \in \Theta_0^c.$$

   (So "power" $= \delta = 1-\beta$.)

5. **Power function and size/level.**
   The **power function** is $\delta(\theta) = \mathbb{P}_\theta(X \in R_0)$ for $\theta \in \Theta_0^c$. A test is a **size α test** if

$$\sup_{\theta \in \Theta_0} \mathbb{P}_\theta(X \in R_0) = \alpha,$$

   and a **level α test** if

$$\sup_{\theta \in \Theta_0} \mathbb{P}_\theta(X \in R_0) \leq \alpha.$$

6. **UMP in a class $\mathcal{C}$.**
   Let $\mathcal{C}$ be a class of tests for $H_0 : \theta \in \Theta_0$ vs $H_1 : \theta \in \Theta_0^c$. A test in $\mathcal{C}$ with power function $\delta(\theta)$ is **uniformly most powerful (UMP) in class $\mathcal{C}$** if

$$\delta(\theta) \geq \delta'(\theta) \quad \text{for every } \theta \in \Theta_0^c,$$

for every other test in $\mathcal{C}$ with power function $\delta'(\theta)$.

Two classical design principles:

$$\text{fix } \alpha \text{ and find } R_0 \text{ that minimizes } \beta, \qquad \text{fix } \beta \text{ and find } R_0 \text{ that minimizes } \alpha.$$

7. **Binomial/Bernoulli example and the α–β trade-off.**
Consider $n = 30$ Bernoulli trials (coin flips) $X_i \sim \text{Ber}(\theta)$ and

$$H_0 : \theta = \frac{1}{2} \quad \text{vs} \quad H_1 : \theta = \frac{2}{3}.$$

Let

$$T = \sum_{i=1}^{30} X_i \sim \text{Binomial}(30, \theta).$$

Consider two rejection regions:

$$R_0^{(1)} : \text{ reject } H_0 \text{ if } T > 25, \qquad R_0^{(2)} : \text{ reject } H_0 \text{ if } T > 17.$$

For $R_0^{(1)}$:

$$\alpha^{(1)} = \mathbb{P}_{\theta=1/2}(T > 25) = \sum_{i=26}^{30} \binom{30}{i} \left(\tfrac{1}{2}\right)^{30} = 2.9738 \times 10^{-5} \approx 0.0000297,$$

$$\beta^{(1)} = \mathbb{P}_{\theta=2/3}(T \le 25) = \sum_{i=0}^{25} \binom{30}{i} \left(\tfrac{2}{3}\right)^{i} \left(\tfrac{1}{3}\right)^{30-i} \approx 0.98777.$$

For $R_0^{(2)}$:

$$\alpha^{(2)} = \mathbb{P}_{\theta=1/2}(T > 17) \approx 0.18080, \qquad \beta^{(2)} = \mathbb{P}_{\theta=2/3}(T \le 17) \approx 0.16601.$$

Inclusion and trade-off. If $R_0^{(1)} \subseteq R_0^{(2)}$, then $\alpha^{(1)} < \alpha^{(2)}$ and typically $\beta^{(1)} > \beta^{(2)}$.

8. **Trade-off proof via inclusion of rejection regions (as in notes).**
Fix two rejection regions R_0' and R_0 (same statistic) with $R_0' \subset R_0$. Then, for every $\theta \in \Theta_0$,

$$\mathbb{P}_\theta(T_n \in R_0') \le \mathbb{P}_\theta(T_n \in R_0),$$

so the Type I error (and hence the size/level bound) cannot increase when shrinking the rejection region. Moreover, for every $\theta \in \Theta_0^c$, since $(R_0')^c \supset R_0^c$,

$$\mathbb{P}_\theta(T_n \notin R_0') = \mathbb{P}_\theta(T_n \in (R_0')^c) \ge \mathbb{P}_\theta(T_n \in R_0^c) = \mathbb{P}_\theta(T_n \notin R_0).$$

Hence reducing Type I error by shrinking R_0 increases Type II error (and reduces power), because the acceptance region is the complement of the rejection region.

Q. Neyman–Pearson (NP) Lemma, Consequences, and One-Parameter Structure

1. **Neyman–Pearson lemma (simple vs. simple) + proof (both directions).**
Consider testing

$$H_0 : \theta = \theta_0 \qquad \text{vs} \qquad H_1 : \theta = \theta_1,$$

and denote by $f_X(x;\theta)$ the density/pmf of X under θ. A (nonrandomized) *test function* is $\phi : \mathcal{X} \to \{0, 1\}$, with $\phi(x) = 1$ meaning "reject H_0".
Define a rejection region R and $\phi = \mathbf{1}_R$ such that, for some $k \ge 0$,

(i) $x \in R$ if $f_X(x;\theta_1) > k\, f_X(x;\theta_0)$ and $x \in R^c$ if $f_X(x;\theta_1) < k\, f_X(x;\theta_0)$;

(ii) $\alpha = \mathbb{P}_{\theta_0}(X \in R)$ (calibration to level α, hence ϕ is a size α test).

Then any test satisfying **(i)–(ii)** is **most powerful** among all level-α tests (and hence "UMP" here since H_1 is pointwise). Moreover, if there exists such a test with $k > 0$, then every UMP level-α test is necessarily a **size** α test; and any two UMP tests coincide except possibly on

$$A := \{x : \ f_X(x;\theta_1) = k f_X(x;\theta_0)\},$$

(in particular, they are a.s. equal under θ_0 and θ_1 if $\mathbb{P}_{\theta_0}(A) = \mathbb{P}_{\theta_1}(A) = 0$).

Proof. Let $\phi = \mathbf{1}_R$ be a test satisfying **(i)–(ii)** and let $\phi'(x) \in \{0,1\}$ be the test function of an arbitrary *level-α* test. Define power functions

$$\delta(\theta) = \mathbb{E}_\theta[\phi(X)], \qquad \delta'(\theta) = \mathbb{E}_\theta[\phi'(X)].$$

($\Rightarrow$) If ϕ satisfies (i)–(ii), then ϕ is MP/UMP level α. By construction of R, for all x,

$$[\phi(x) - \phi'(x)]\big(f_X(x;\theta_1) - k f_X(x;\theta_0)\big) \geq 0,$$

indeed:

- if $f_X(x;\theta_1) > k f_X(x;\theta_0)$ then $x \in R$ hence $\phi(x) = 1$;
- if $\phi(x) = 0$ then $x \notin R$ hence $f_X(x;\theta_1) < k f_X(x;\theta_0)$.

Integrating over $\mathcal{X}$ (write dx for brevity),

$$\begin{aligned} 0 &\leq \int_{\mathcal{X}} [\phi(x) - \phi'(x)]\big(f_X(x;\theta_1) - k f_X(x;\theta_0)\big)\, dx \\ &= \delta(\theta_1) - \delta'(\theta_1) - k\big(\delta(\theta_0) - \delta'(\theta_0)\big). \end{aligned}$$

Since $\delta(\theta_0) = \alpha$ (size of ϕ) and $\delta'(\theta_0) \leq \alpha$ (level of ϕ'), we have

$$\delta(\theta_0) - \delta'(\theta_0) = \alpha - \delta'(\theta_0) \geq 0.$$

With $k \geq 0$,

$$0 \leq \delta(\theta_1) - \delta'(\theta_1) - k(\alpha - \delta'(\theta_0)) \leq \delta(\theta_1) - \delta'(\theta_1),$$

hence $\delta(\theta_1) \geq \delta'(\theta_1)$ for every level-α test ϕ'. Thus ϕ is most powerful at θ_1.

($\Leftarrow$) Conversely: if ϕ' is UMP level α, then (when $k > 0$) ϕ' must be size α. Assume there exists a test ϕ satisfying **(i)–(ii)** with $k > 0$, and let ϕ' be the test function of any UMP level-α test. Since ϕ is MP level α by the first part and ϕ' is also UMP (hence MP) level α, they must have the same power at θ_1:

$$\delta(\theta_1) = \delta'(\theta_1).$$

From the integrated inequality,

$$0 \leq \delta(\theta_1) - \delta'(\theta_1) - k\big(\delta(\theta_0) - \delta'(\theta_0)\big),$$

substituting $\delta(\theta_1) = \delta'(\theta_1)$ gives

$$0 \leq -k\big(\delta(\theta_0) - \delta'(\theta_0)\big).$$

Since $k > 0$, dividing by $-k$ reverses the inequality:

$$\delta(\theta_0) - \delta'(\theta_0) \leq 0.$$

Using $\delta(\theta_0) = \alpha$ (because ϕ is size α) yields

$$\alpha - \delta'(\theta_0) \leq 0 \quad \Longrightarrow \quad \alpha \leq \delta'(\theta_0).$$

But ϕ' is level α, so $\delta'(\theta_0) \leq \alpha$. Hence

$$\delta'(\theta_0) = \alpha,$$

i.e. ϕ' is a *size* α test.

(Uniqueness up to the boundary). The pointwise inequality shows that ϕ and ϕ' can disagree only on the boundary

$$A = \{x : \ f_X(x;\theta_1) = k f_X(x;\theta_0)\}.$$

In particular, if $\mathbb{P}_{\theta_0}(A) = \mathbb{P}_{\theta_1}(A) = 0$, then $\phi = \phi'$ $\mathbb{P}_{\theta_0}$-a.s. and $\mathbb{P}_{\theta_1}$-a.s. $\square$

2. NP examples: Normal mean (variance known) and Bernoulli (discrete calibration).

(a) Normal mean, variance known (one-sided).

Let $X_1, \dots, X_n$ be i.i.d. $N(\theta, \sigma^2)$ with σ^2 known. Test

$$H_0 : \theta = \theta_0 \qquad \text{vs} \qquad H_1 : \theta = \theta_1 > \theta_0.$$

The likelihood is

$$L(\theta; x) = \prod_{i=1}^{n} \frac{1}{\sqrt{2\pi}\sigma} \exp\left\{ -\frac{1}{2\sigma^2}(x_i - \theta)^2 \right\} = \left(\frac{1}{\sqrt{2\pi}\sigma} \right)^n \exp\left\{ -\frac{1}{2\sigma^2} \sum_{i=1}^{n} (x_i - \theta)^2 \right\}.$$

Hence the likelihood ratio is

$$\begin{aligned}
\Lambda(x) = \frac{L(\theta_1; x)}{L(\theta_0; x)} &= \exp\left\{ -\frac{1}{2\sigma^2} \left[\sum_{i=1}^{n} (x_i - \theta_1)^2 - \sum_{i=1}^{n} (x_i - \theta_0)^2 \right] \right\} \\
&= \exp\left\{ -\frac{1}{2\sigma^2} \sum_{i=1}^{n} \left[(x_i^2 - 2\theta_1 x_i + \theta_1^2) - (x_i^2 - 2\theta_0 x_i + \theta_0^2) \right] \right\} \\
&= \exp\left\{ -\frac{1}{2\sigma^2} \left[-2(\theta_1 - \theta_0) \sum_{i=1}^{n} x_i + n(\theta_1^2 - \theta_0^2) \right] \right\} \\
&= \exp\left\{ \frac{\theta_1 - \theta_0}{\sigma^2} \sum_{i=1}^{n} x_i - \frac{n(\theta_1^2 - \theta_0^2)}{2\sigma^2} \right\} \\
&= \exp\left\{ \frac{n(\theta_1 - \theta_0)}{\sigma^2} \left(\bar{x} - \frac{\theta_1 + \theta_0}{2} \right) \right\}.
\end{aligned}$$

Since $\theta_1 > \theta_0$, $\Lambda(x)$ is strictly increasing in $\bar{x}$. Therefore the NP rejection region $\{\Lambda(x) \geq c\}$ is equivalent to a one-sided threshold on the sample mean:

$$\Lambda(x) \geq c \iff \frac{n(\theta_1 - \theta_0)}{\sigma^2} \left(\bar{x} - \frac{\theta_1 + \theta_0}{2} \right) \geq \log c \iff \bar{x} \geq k,$$

where

$$k = \frac{\theta_1 + \theta_0}{2} + \frac{\sigma^2}{n(\theta_1 - \theta_0)} \log c.$$

To get a size-α test, choose k so that

$$\alpha = \mathbb{P}_{\theta_0}(\bar{X} \geq k).$$

Under H_0, $\bar{X} \sim N\left(\theta_0, \frac{\sigma^2}{n}\right)$, hence

$$\alpha = \mathbb{P}_{\theta_0}\left(\frac{\bar{X} - \theta_0}{\sigma/\sqrt{n}} \geq \frac{k - \theta_0}{\sigma/\sqrt{n}} \right) = 1 - \Phi\left(\frac{k - \theta_0}{\sigma/\sqrt{n}} \right).$$

Let $z_{1-\alpha} = \Phi^{-1}(1 - \alpha)$ (equivalently, define z_α by $1 - \Phi(z_\alpha) = \alpha$). Then

$$\frac{k - \theta_0}{\sigma/\sqrt{n}} = z_{1-\alpha} \qquad \Longrightarrow \qquad k = \theta_0 + z_{1-\alpha} \frac{\sigma}{\sqrt{n}}.$$

Therefore the NP (MP, and here UMP) level-α test rejects H_0 when

$$R = \left\{ \bar{X} \geq \theta_0 + z_{1-\alpha} \frac{\sigma}{\sqrt{n}} \right\} \qquad \left(\text{equivalently } R = \{ \bar{X} \geq \theta_0 + z_\alpha \tfrac{\sigma}{\sqrt{n}} \} \text{ with } 1 - \Phi(z_\alpha) = \alpha \right).$$

(b) Bernoulli (one-sided) and the discrete issue.
Let $X_1, \dots, X_n$ be i.i.d. $\mathrm{Ber}(\theta)$ and consider

$$H_0 : \theta = \theta_0 \qquad \text{vs} \qquad H_1 : \theta = \theta_1 > \theta_0.$$

Write $T = \sum_{i=1}^n X_i$ (so $T \sim \mathrm{Bin}(n, \theta)$). The likelihood is

$$L(\theta; x) = \prod_{i=1}^n \theta^{x_i}(1-\theta)^{1-x_i} = \theta^{\sum_{i=1}^n x_i}(1-\theta)^{n-\sum_{i=1}^n x_i} = \theta^T(1-\theta)^{n-T}.$$

Hence the likelihood ratio is

$$\begin{aligned}
\Lambda(x) = \frac{L(\theta_1; x)}{L(\theta_0; x)} &= \left(\frac{\theta_1}{\theta_0}\right)^T \left(\frac{1-\theta_1}{1-\theta_0}\right)^{n-T}, \\
\log \Lambda(x) &= T \log\left(\frac{\theta_1}{\theta_0}\right) + (n-T)\log\left(\frac{1-\theta_1}{1-\theta_0}\right) \\
&= n \log\left(\frac{1-\theta_1}{1-\theta_0}\right) + T\left[\log\left(\frac{\theta_1}{\theta_0}\right) - \log\left(\frac{1-\theta_1}{1-\theta_0}\right)\right].
\end{aligned}$$

If $\theta_1 > \theta_0$, then

$$\frac{\theta_1}{\theta_0} > 1 \;\Rightarrow\; \log\left(\frac{\theta_1}{\theta_0}\right) > 0, \qquad \frac{1-\theta_1}{1-\theta_0} < 1 \;\Rightarrow\; \log\left(\frac{1-\theta_1}{1-\theta_0}\right) < 0,$$

so the bracketed coefficient of T is strictly positive and therefore $\Lambda(x)$ is strictly increasing in T. Thus the Neyman–Pearson rejection region $\{\Lambda(x) \geq k\}$ is equivalent to a threshold test:

$$R = \{T \geq c\} \quad \text{for some integer } c \in \{0, 1, \dots, n\}.$$

Calibration (discrete issue). A size-α calibration would require

$$\mathbb{P}_{\theta_0}(T \geq c) = \alpha, \qquad T \sim \mathrm{Bin}(n, \theta_0),$$

but since T is discrete, an integer c with equality may not exist. Typically one chooses

$$c = \min\left\{m \in \{0, \dots, n\} : \; \mathbb{P}_{\theta_0}(T \geq m) \leq \alpha\right\},$$

which guarantees a *level-α* test (size may be $< \alpha$):

$$\mathbb{P}_{\theta_0}(T \geq c) \leq \alpha.$$

Exact size via randomization (optional). If one allows randomized tests, one can achieve exact size α by rejecting when $T > c$ and rejecting when $T = c$ with probability $\gamma \in [0, 1]$ chosen so that

$$\mathbb{P}_{\theta_0}(T > c) + \gamma\, \mathbb{P}_{\theta_0}(T = c) = \alpha.$$

Numerical example. Let $n = 20$, $\theta_0 = 0.3$, $\alpha = 0.05$. Taking $c = 10$ gives

$$\mathbb{P}_{\theta_0}(T \geq 10) = \sum_{t=10}^{20} \binom{20}{t} (0.3)^t (0.7)^{20-t} = 0.04796 < 0.05.$$

Asymptotic note. As $n \to \infty$, the binomial tail probabilities become effectively continuous (e.g. via CLT/normal approximation), so matching a target size becomes easier in practice.

3. One-parameter exponential family: LR shape and threshold in $\sum A(X_i)$.
Assume a one-parameter exponential family with density/pmf

$$f_X(x;\theta) = \exp\{A(x)Q(\theta) + C(x) - K(\theta)\}, \qquad \theta \in \Theta \subset \mathbb{R}.$$

For an i.i.d. sample $X_1, \ldots, X_n$, the likelihood is

$$\begin{aligned} L(\theta; X) &= \prod_{i=1}^{n} f_X(X_i;\theta) \\ &= \prod_{i=1}^{n} \exp\{A(X_i)Q(\theta) + C(X_i) - K(\theta)\} \\ &= \exp\left\{ Q(\theta) \sum_{i=1}^{n} A(X_i) + \sum_{i=1}^{n} C(X_i) - nK(\theta) \right\}. \end{aligned}$$

Consider testing

$$H_0 : \theta = \theta_0 \qquad \text{vs} \qquad H_1 : \theta = \theta_1 > \theta_0,$$

and the Neyman–Pearson rejection rule $\frac{L(\theta_1;X)}{L(\theta_0;X)} \geq k$ for some $k > 0$. Compute the likelihood ratio:

$$\begin{aligned} \frac{L(\theta_1; X)}{L(\theta_0; X)} &= \frac{\exp\left\{Q(\theta_1)\sum_{i=1}^{n} A(X_i) + \sum_{i=1}^{n} C(X_i) - nK(\theta_1)\right\}}{\exp\left\{Q(\theta_0)\sum_{i=1}^{n} A(X_i) + \sum_{i=1}^{n} C(X_i) - nK(\theta_0)\right\}} \\ &= \exp\left\{ \big(Q(\theta_1) - Q(\theta_0)\big) \sum_{i=1}^{n} A(X_i) - n\big(K(\theta_1) - K(\theta_0)\big) \right\}. \end{aligned}$$

Thus

$$\begin{aligned} \frac{L(\theta_1; X)}{L(\theta_0; X)} \geq k &\iff \big(Q(\theta_1) - Q(\theta_0)\big) \sum_{i=1}^{n} A(X_i) - n\big(K(\theta_1) - K(\theta_0)\big) \geq \log k \\ &\iff \big(Q(\theta_1) - Q(\theta_0)\big) \sum_{i=1}^{n} A(X_i) \geq \log k + n\big(K(\theta_1) - K(\theta_0)\big). \end{aligned}$$

Let

$$c := \frac{\log k + n\big(K(\theta_1) - K(\theta_0)\big)}{Q(\theta_1) - Q(\theta_0)}.$$

Then the rejection region can be written as a threshold rule in the statistic

$$T = \sum_{i=1}^{n} A(X_i) : \qquad R = \begin{cases} \{\, T \geq c \,\}, & \text{if } Q(\theta_1) - Q(\theta_0) > 0, \\ \{\, T \leq c \,\}, & \text{if } Q(\theta_1) - Q(\theta_0) < 0. \end{cases}$$

Equivalently, the LR is monotone in T, so the NP test rejects for large (or small) values of T depending on the sign of $Q(\theta_1) - Q(\theta_0)$.
Calibration. Choose c (equivalently k) so that under H_0

$$\mathbb{P}_{\theta_0}(T \in R) = \alpha \quad (\text{or } \leq \alpha \text{ in discrete models, unless randomization is allowed}).$$

Remark. The statistic $T = \sum_{i=1}^{n} A(X_i)$ is sufficient for θ in a one-parameter EF, and the NP rejection region depends on the sample only through T.

4. Corollary: Neyman–Pearson with a sufficient statistic (factorization consequence).
Consider the NP setting

$$H_0 : \theta = \theta_0 \qquad \text{vs} \qquad H_1 : \theta = \theta_1,$$

and let $T = T(X)$ be sufficient for θ. Denote by $g_T(t;\theta)$ the density/pmf of T under θ.

If the model admits the Fisher–Neyman factorization

$$f_X(x;\theta) = h(x)\, g_T(T(x);\theta),$$

then the likelihood ratio satisfies

$$\frac{f_X(x;\theta_1)}{f_X(x;\theta_0)} = \frac{h(x)\, g_T(T(x);\theta_1)}{h(x)\, g_T(T(x);\theta_0)} = \frac{g_T(T(x);\theta_1)}{g_T(T(x);\theta_0)}.$$

Therefore the NP rejection region $\{f_X(x;\theta_1) > k f_X(x;\theta_0)\}$ can be written in terms of T only:

$$S := \{t :\ g_T(t;\theta_1) > k\, g_T(t;\theta_0)\}, \qquad R = \{x :\ T(x) \in S\}.$$

Choose $k \geq 0$ (equivalently, a boundary for S) so that the test has level α:

$$\alpha = \mathbb{P}_{\theta_0}(T \in S).$$

Then the test $\phi(x) = \mathbf{1}\{T(x) \in S\}$ is most powerful among all level-α tests; since H_1 is simple, it is also UMP in this setting.

5. **Rubin (MLR) corollary for one-sided composite UMP tests.**
MLR definition. A family $\{g_T(t;\theta) : \theta \in \Theta\}$ has the *monotone likelihood ratio* (MLR) property in t if for every $\theta_2 > \theta_1$ the ratio

$$\frac{g_T(t;\theta_2)}{g_T(t;\theta_1)}$$

is a monotone function of t (non-decreasing or non-increasing).
Consider the one-sided composite problem

$$H_0 : \theta \leq \theta_0 \qquad \text{vs} \qquad H_1 : \theta > \theta_0.$$

Assume:
(i) $T = T(X)$ is sufficient for θ;
(ii) the family $\{g_T(t;\theta) : \theta \in \Theta\}$ has MLR in t.
Conclusion (Rubin/Karlin–Rubin). If $\frac{g_T(t;\theta_2)}{g_T(t;\theta_1)}$ is non-decreasing in t, then for each $\alpha \in (0,1)$ the test

$$\phi(x) = \mathbf{1}\{T(x) > t_\alpha\}$$

is UMP among all level-α tests for $H_0 : \theta \leq \theta_0$ vs $H_1 : \theta > \theta_0$, where t_α is chosen so that

$$\mathbb{P}_{\theta_0}(T > t_\alpha) = \alpha \quad (\text{or } \leq \alpha \text{ in the discrete case, unless randomization is allowed}).$$

If instead the ratio is non-increasing in t, then the UMP test rejects for small values:

$$\phi(x) = \mathbf{1}\{T(x) < t_\alpha\}.$$

Remark. In one-parameter exponential families, $T = \sum_{i=1}^n A(X_i)$ is sufficient and the MLR property typically holds, yielding one-sided UMP tests of threshold form.

R. Likelihood Ratio Test (LRT): Finite-Sample Forms and Examples

1. **Definition of the LRT.** Consider testing

$$H_0 :\ \theta \in \Theta_0 \subseteq \Theta \qquad \text{vs} \qquad H_1 :\ \theta \in \Theta \setminus \Theta_0.$$

Given the likelihood $L(\theta; x) = L(\theta; x_1, \ldots, x_n)$, define the likelihood ratio statistic

$$\Lambda(x_1, \ldots, x_n) = \frac{\sup_{\theta \in \Theta_0} L(\theta; x)}{\sup_{\theta \in \Theta} L(\theta; x)} \in (0, 1].$$

The LRT rejects H_0 for small values of Λ, i.e.

$$R = \{x : \Lambda(x) \le c\}, \qquad c \in (0,1),$$

where c is chosen so that $\mathbb{P}_{\theta\in\Theta_0}(\Lambda(X) \le c) \le \alpha$ (and equals α when the null distribution is continuous or randomization is allowed).

2. **Classical finite-sample LRTs in Gaussian models (exact reductions).**

(a) **One-sample normal, unknown mean and variance: LRT $\Rightarrow$ two-sided t-test.**
Model:

$$X_1, \dots, X_n \text{ iid } \sim \mathcal{N}(\theta, \sigma^2), \qquad \theta \in \mathbb{R},\ \sigma^2 > 0.$$

Test:

$$H_0 : \theta = \theta_0 \qquad \text{vs} \qquad H_1 : \theta \neq \theta_0.$$

Unrestricted MLEs:

$$\widehat{\theta} = \bar{X}, \qquad \widehat{\sigma}^2 = \frac{1}{n}\sum_{i=1}^{n}(X_i - \bar{X})^2.$$

Under H_0 (mean fixed at θ_0), the MLE of σ^2 is

$$\widehat{\sigma}_0^2 = \frac{1}{n}\sum_{i=1}^{n}(X_i - \theta_0)^2.$$

The LRT statistic is

$$\Lambda = \left(\frac{\widehat{\sigma}^2}{\widehat{\sigma}_0^2}\right)^{n/2}.$$

Using the identity

$$\widehat{\sigma}_0^2 = \widehat{\sigma}^2 + (\bar{X} - \theta_0)^2,$$

and the corrected sample variance

$$s^2 = \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \bar{X})^2, \qquad \widehat{\sigma}^2 = \frac{n-1}{n}s^2,$$

the rejection region $\{\Lambda \le c\}$ is equivalent to a threshold on

$$T_n = \frac{\sqrt{n}\,|\bar{X} - \theta_0|}{s}.$$

Under H_0, $T_n \sim t_{n-1}$. Hence the LRT coincides with the usual two-sided t-test: reject H_0 at level α when

$$|T_n| \ge t_{n-1,\,1-\alpha/2}.$$

(b) **Two-sample normal, equal variances: LRT for equality of means $\Rightarrow$ pooled two-sample t-test.**
Model (independent samples):

$$(X_1, \dots, X_\mu) \sim \mathcal{N}(\theta_x, \sigma^2), \qquad (Y_1, \dots, Y_\nu) \sim \mathcal{N}(\theta_y, \sigma^2), \qquad X \perp Y.$$

Test:

$$H_0 : \theta_x = \theta_y \qquad \text{vs} \qquad H_1 : \theta_x \neq \theta_y.$$

Unrestricted MLEs:

$$\widehat{\theta}_x = \bar{X}_\mu, \qquad \widehat{\theta}_y = \bar{Y}_\nu,$$

and the unrestricted variance MLE

$$\widehat{S}^2 = \frac{1}{\mu+\nu}\left[\sum_{i=1}^{\mu}(X_i - \bar{X}_\mu)^2 + \sum_{j=1}^{\nu}(Y_j - \bar{Y}_\nu)^2\right].$$

Under H_0 the common-mean MLE is

$$\theta_0^* = \frac{\mu \bar{X}_\mu + \nu \bar{Y}_\nu}{\mu + \nu},$$

and the corresponding variance MLE is

$$\widetilde{S}_0^2 = \frac{1}{\mu + \nu} \left[\sum_{i=1}^{\mu} (X_i - \theta_0^*)^2 + \sum_{j=1}^{\nu} (Y_j - \theta_0^*)^2 \right].$$

The LRT statistic is

$$\Lambda = \left(\frac{\widehat{S}^2}{\widetilde{S}_0^2} \right)^{(\mu+\nu)/2}.$$

The rejection region $\{\Lambda \le c\}$ is equivalent to rejecting for large values of the pooled two-sample t statistic

$$T_{\mu,\nu} = \frac{\bar{X}_\mu - \bar{Y}_\nu}{S_p \sqrt{\frac{1}{\mu} + \frac{1}{\nu}}}, \qquad S_p^2 = \frac{\sum_{i=1}^{\mu} (X_i - \bar{X}_\mu)^2 + \sum_{j=1}^{\nu} (Y_j - \bar{Y}_\nu)^2}{\mu + \nu - 2}.$$

Under H_0, $T_{\mu,\nu} \sim t_{\mu+\nu-2}$. Hence the LRT reduces exactly to the two-sided pooled t-test: reject for $|T_{\mu,\nu}|$ large, i.e.

$$|T_{\mu,\nu}| \ge t_{\mu+\nu-2,\, 1-\alpha/2}.$$

(c) **Two-sample normal: LRT for equality of variances $\Rightarrow$ F-test.**
Model:

$$(X_1, \ldots, X_\mu) \sim \mathcal{N}(\theta_x, \sigma_x^2), \qquad (Y_1, \ldots, Y_\nu) \sim \mathcal{N}(\theta_y, \sigma_y^2), \qquad X \perp Y.$$

Test:

$$H_0 : \sigma_x^2 = \sigma_y^2 \qquad \text{vs} \qquad H_1 : \sigma_x^2 \ne \sigma_y^2.$$

Let

$$S_X^2 = \frac{1}{\mu - 1} \sum_{i=1}^{\mu} (X_i - \bar{X}_\mu)^2, \qquad S_Y^2 = \frac{1}{\nu - 1} \sum_{j=1}^{\nu} (Y_j - \bar{Y}_\nu)^2.$$

Under H_0 and Gaussianity,

$$\frac{S_X^2}{S_Y^2} \sim F_{\mu-1,\nu-1}.$$

Therefore an exact two-sided level-α test rejects when

$$\frac{S_X^2}{S_Y^2} \le F_{\mu-1,\nu-1;\,\alpha/2} \quad \text{or} \quad \frac{S_X^2}{S_Y^2} \ge F_{\mu-1,\nu-1;\,1-\alpha/2}.$$

3. **Why Gaussianity matters for exact finite-sample $t/F/\chi^2$ pivots.** The exact reductions above rely on normal-theory identities: centered sums of squares of normals are χ^2, the relevant quadratic forms are independent, and ratios yield exact t and F distributions. Without Gaussianity these pivot laws generally fail in finite samples, and one typically replaces exact calibration with asymptotic theory (e.g. Wilks) or robust/large-sample methods.

S. Asymptotic LRT: Wilks Theorem

The likelihood ratio test (LRT) is used to build a rejection region (without guaranteeing optimality). We test

$$H_0 : \ \theta \in \Theta_0 \subseteq \Theta \qquad \text{vs} \qquad H_1 : \ \theta \notin \Theta_0,$$

using the likelihood ratio statistic

$$\Lambda(x_1, \ldots, x_u) = \frac{\sup_{\theta \in \Theta_0} L(\theta; x)}{\sup_{\theta \in \Theta} L(\theta; x)}.$$

A typical rejection rule is to reject H_0 for small values of Λ, i.e.

$$\text{Reject } H_0 \text{ if } \Lambda(x_1, \ldots, x_u) \leq c.$$

Example: if $x_1, \ldots, x_u$ are i.i.d. $\mathcal{N}(\theta, \sigma^2)$ with σ^2 known, then

$$H_0 : \ \theta = \theta_0 \quad \text{vs} \quad H_1 : \ \theta \neq \theta_0,$$

and one can express the rejection region in terms of the sum (or equivalently the sample mean), e.g.

$$\text{Reject } H_0 \text{ if } \sum_{i=1}^{u} x_i > c,$$

where c is chosen so that

$$\mathbb{P}\left(\sum_{i=1}^{u} x_i > c \right) = \alpha,$$

which is a quantile problem and requires knowing the distribution of $\sum_{i=1}^{u} x_i$. Closed-form quantiles are rare (e.g. uniform, exponential), so in practice one uses tabulated distributions (χ^2, t, F, Gaussian/standard Gaussian) or asymptotic approximations. Common asymptotic tools are Wilks' theorem and the Wald test.

1. **Wilks' theorem (general form).**
   Let $(x_1, \ldots, x_n)$ be a random sample from a regular parametric model

   $$\{f_\theta : \ \theta \in \Theta\},$$

   where $\Theta \subset \mathbb{R}^p$ is open. Consider testing

   $$H_0 : \ \theta \in \Theta_0 \qquad \text{vs} \qquad H_1 : \ \theta \in \Theta \setminus \Theta_0.$$

   where Θ_0 is defined by r independent smooth constraints (so $\dim(\Theta_0) = p - r$). Define the likelihood ratio statistic

   $$\Lambda(x_1, \ldots, x_n) = \frac{\sup_{\theta \in \Theta_0} L(\theta \,|\, x)}{\sup_{\theta \in \Theta} L(\theta \,|\, x)}.$$

   Then under H_0 and as $n \to \infty$,

   $$-2 \log \Lambda(x_1, \ldots, x_n) \ \xrightarrow{d} \ \chi^2_r.$$

   Hence an asymptotic level-α LRT rejects when

   $$-2 \log \Lambda(x_1, \ldots, x_n) \ \geq \ \chi^2_{r,\,1-\alpha}.$$

   Proof. Let $\ell(\theta; n) := \log L(\theta; x)$ and let $\hat{\theta}_u$ be the MLE.

   $$\log \Lambda = \log L(\theta_0; n) - \log L(\hat{\theta}_u; x) = \ell(\theta_0; n) - \ell(\hat{\theta}_u; x).$$

   $$-2 \log \Lambda = -2\ell(\theta_0; n) + 2\ell(\hat{\theta}_u; n).$$

   $$-2 \log \Lambda = -2\Big[\underbrace{\ell(\hat{\theta}_u; n)}_{A} + \underbrace{\ell'(\hat{\theta}_u; n)(\theta_0 - \hat{\theta}_u)}_{B} + \underbrace{\frac{1}{2}\, \ell''(\hat{\theta}_u; n)(\theta_0 - \hat{\theta}_u)^2}_{C} + \underbrace{r_n}_{D} \Big] + \underbrace{2\ell(\hat{\theta}_u; x)}_{E}.$$

   $$A + E = 0 \quad B = 0 \quad \text{since } \hat{\theta}_u \text{ maximizes } \ell(\theta; x) \text{ so } \ell'(\hat{\theta}_u, x) = 0.$$

   Since we are interested in studying the asymptotic behaviour when $u \to \infty$ the r_u (D) is negligible. Hence

   $$-2 \log \Lambda \ \simeq \ -\ell''(\hat{\theta}_u; x)\, (\theta_0 - \hat{\theta}_u)^2 \qquad \text{when } n \to +\infty.$$

$$-2\log\Lambda \simeq \underbrace{-\frac{1}{u}\,\ell''(\hat\theta_u;x)}_{a}\cdot\underbrace{u(\theta_0-\hat\theta_u)^2}_{b}.$$

$$a = R^2 \xrightarrow[u\to\infty]{} \mathcal{N}\!\left(0,\frac{1}{I(\theta_0)}\right) \qquad \text{for Cramer Theor.}$$

$$b \xrightarrow[u\to\infty]{} I(\theta_0).$$

Hence

$$-2\log\Lambda \xrightarrow[u\to\infty]{} R^2 I(\theta_0) = \left(\sqrt{I(\theta_0)}\,R\right)^2 = Z^2, \qquad Z\sim\mathcal{N}(0,1).$$

So the limit distribution is χ^2_1 (and in general constrained case χ^2_r).

□

2. **Poisson LRT example.** Let $(X_1,\dots,X_n)$ be a random sample from $X\sim\text{Poisson}(\lambda)$, $\lambda>0$, and test

$$H_0:\lambda=\lambda_0 \qquad \text{vs} \qquad H_1:\lambda\neq\lambda_0.$$

The likelihood is

$$L(\lambda;x)=\prod_{i=1}^n \frac{e^{-\lambda}\lambda^{x_i}}{x_i!}=\frac{e^{-n\lambda}\lambda^{\sum_{i=1}^n x_i}}{\prod_{i=1}^n x_i!}.$$

The unrestricted MLE is $\widehat\lambda_n=\bar X_n$. The LR statistic is

$$\Lambda=\frac{L(\lambda_0;x)}{L(\widehat\lambda_n;x)}=\exp\big(-n(\lambda_0-\widehat\lambda_n)\big)\left(\frac{\lambda_0}{\widehat\lambda_n}\right)^{\sum_{i=1}^n x_i}.$$

Using $\sum_{i=1}^n x_i = n\widehat\lambda_n$, we obtain

$$-2\log\Lambda = 2n\left[\widehat\lambda_n\log\left(\frac{\widehat\lambda_n}{\lambda_0}\right)-(\widehat\lambda_n-\lambda_0)\right].$$

Under H_0, Wilks theorem implies

$$-2\log\Lambda \xrightarrow{d} \chi^2_1 \qquad (n\to\infty),$$

By using table of χ^2 an asymptotic level-α test rejects when $-2\log\Lambda\geq\chi^2_{1,1-\alpha}$.

3. **Cautionary example (non-regular model: Wilks may fail).** Let $X_1,\dots,X_n$ be i.i.d. with density

$$f(x;\theta)=e^{-(x-\theta)}\mathbf{1}\{x\geq\theta\}, \qquad \theta\in\mathbb{R}.$$

Testing $H_0:\theta=\theta_0$ vs $H_1:\theta\neq\theta_0$, the likelihood is proportional to $e^{-\sum_{i=1}^n(x_i-\theta)}\mathbf{1}\{\theta\leq X_{(1)}\}$, hence the MLE is $\widehat\theta_n=X_{(1)}$. One finds

$$-2\log\Lambda = 2n(\widehat\theta_n-\theta_0)=2n(X_{(1)}-\theta_0),$$

and under H_0, $n(X_{(1)}-\theta_0)\sim\text{Exp}(1)$, so

$$-2\log\Lambda\sim\text{Exp}\left(\tfrac{1}{2}\right),$$

which is *not* a χ^2 distribution. (This happens because the support depends on θ and the model is not regular.)

4. **Theorem (Multidimensional Wilks).** Let $(x_1,\dots,x_n)$ be a random sample from a regular model $X\sim F_\theta$, $\theta\in\Theta$, with Θ that can have any finite dimension. If $\theta\in\Theta_0$, then the distribution of

$$-2\log\Lambda(x_1,\dots,x_n)$$

converges to the distribution of a chi-squared random variable as $n \to \infty$. The degrees of freedom of the limiting chi-squared distribution is the difference between the number of free parameters specified by the null $H_0 : \theta \in \Theta_0$ and the number of free parameters specified by $\theta \in \Theta$ (full model, i.e. free from constraints).

$$(\theta_0 - \hat{\theta}_n)^2, \qquad \sqrt{n}\,(\theta_0 - \hat{\theta}_n) \;\to\; N(.).$$

Example: linear model.

$$X \in \mathbb{R}^{n\times p}, \qquad \beta \in \mathbb{R}^p, \qquad \varepsilon \in \mathbb{R}^n,$$
$$y = X\beta + \varepsilon, \qquad \varepsilon \sim \mathcal{N}(0, \sigma^2 I_n) \quad \Longrightarrow \quad y \sim \mathcal{N}(X\beta, \sigma^2 I_n).$$

5. **Example (multidimensional version of the experiment of flipping a coin).**
Let $\theta = (p_1, \ldots, p_s)$ where $p_j \in (0,1)$, $j = 1, \ldots, s$, and $\sum_{j=1}^s p_j = 1$ (constraint). Let $(x_1, \ldots, x_n)$ be a random sample such that $\Pr[X_i = j] = p_j,\ j = 1, \ldots, 5$. (In the notes: $X_i \in \{1, 2, 3, 4, 5\}$.) Define $y_j = \sum_{i=1}^u \mathbf{1}\{x_i = j\}$, the number of times I get j in n attempts, so $y_j \in \{0, \ldots, n\}$ and $\sum_{j=1}^5 y_j = 1$. Then the likelihood is

$$L(\theta; x) = \Pr\left(\langle Y_1 = y_1\rangle \cap \cdots \cap \langle Y_5 = y_5\rangle\right) = p_1^{y_1} \cdots p_5^{y_5}\, \frac{n!}{y_1! \cdots y_5!},$$

where $\dfrac{n!}{y_1! \cdots y_5!}$ is *equivalent to a binomial coefficient.*
Using $p_5 = 1 - \sum_{j=1}^4 p_j$ and $y_5 = n - \sum_{j=1}^4 y_j$:

$$L(\theta; x) = \left(\prod_{j=1}^4 p_j^{y_j}\right)\left(1 - \sum_{j=1}^4 p_j\right)^{n - \sum_{j=1}^4 y_j} \frac{n!}{\left(\prod_{j=1}^4 y_j!\right)\left(n - \sum_{j=1}^4 y_j\right)!}.$$

Hypotheses.

$$H_0 :\ p_1 = p_2 = p_3 \ \wedge\ p_4 = p_5 \qquad \text{vs} \qquad H_1 :\ H_0 \text{ is false} \quad \Rightarrow \quad \text{apply the LRT.}$$

Unrestricted MLE.

$$\frac{\partial}{\partial p_j} \log L(\theta; x) = 0, \qquad j = 1, \ldots, 4,$$
$$\hat{p}_j = \frac{y_j}{n}, \qquad j = 1, \ldots, 4, \qquad \hat{p}_5 = 1 - \sum_{j=1}^4 \hat{p}_j.$$

MLE under H_0. Under H_0, $p_1 = p_2 = p_3$ and $p_4 = p_5$, and with $\sum_{j=1}^5 p_j = 1$:

$$p_1 + p_2 + p_3 + p_4 + p_5 = 1 \quad \Longrightarrow \quad 3p_1 + (p_4 + p_5) = 1, \qquad p_4 = p_5 = \frac{1 - 3p_1}{2}.$$

Thus (up to constants not depending on p_1)

$$L^{(0)}(\theta; x) = p_1^{y_1 + y_2 + y_3}\left(\frac{1 - 3p_1}{2}\right)^{y_4 + y_5}.$$

The constrained MLE solves

$$\frac{d}{dp_1} \log L^{(0)}(\theta; x) = 0 \quad \Longrightarrow \quad \hat{p}_1^{(0)} = \frac{y_1 + y_2 + y_3}{3n} = \hat{p}_2^{(0)} = \hat{p}_3^{(0)},$$

and

$$\hat{p}_4^{(0)} = \hat{p}_5^{(0)} = \frac{1 - 3\hat{p}_1^{(0)}}{2}.$$

Likelihood xstatistic.

$$\Lambda(x_1,\ldots,x_u)=\frac{\sup_{\Theta_0}L(\theta;x)}{\sup_{\Theta}L(\theta;x)}=\left(\frac{y_1+y_2+y_3}{3n}\right)^{y_1}\left(\frac{y_1+y_2+y_3}{3n}\right)^{y_2}\left(\frac{y_1+y_2+y_3}{3n}\right)^{y_3}\cdot\left(\frac{y_4+y_5}{2n}\right)^{y_4}\left(\frac{y_4+y_5}{2n}\right)^{y_5}.$$

$$\Pr[\Lambda(x)\le c]=\alpha. \quad\Rightarrow\quad \text{apply Wilks} \qquad \text{(notes: "find the distribution is impossible").}$$

$$-2\log\Lambda(x_1,\ldots,x_n)\ \xrightarrow{w}\ H, \qquad n\to\infty, \qquad H\sim\chi^2_{4-1}.$$

(Notes on degrees of freedom: full model has 5 parameters with 1 constraint $\sum_{j=1}^5 p_j=1\Rightarrow 4$ free parameters; under H_0 there are 2 parameters with 1 constraint $\Rightarrow 1$ free parameter.)

6. **Wald test.** Based on the use of an estimator T_n for the parameter $\theta\in\Theta$, where T_n must have a gaussian limiting distribution, with $(x_1,\ldots,x_n)$ from F_θ, $\theta\in\Theta$.
We need an estimator T_n such that as $n\to\infty$ the distribution of T_n becomes gaussian. Denote by

$$T_n:=T(x_1,\ldots,x_n)$$

a point estimator for a real-valued parameter $\theta\in\Theta$ (dimension 1).
If σ_n^2 is the variance of estimator T_n and we can apply the CLT (Lindeberg) to show that

$$\frac{T_n-\theta}{s_n}\ \xrightarrow{w}\ Z\sim\mathcal{N}(0,1),$$

then we can compare the statistic $\dfrac{T_n-\theta}{\sigma_n}$ with $Z\sim\mathcal{N}(0,1)$. This is the basis to construct a test for θ (asymptotic).

7. **Consistency**.

Let $(x_1,\ldots,x_n)$ be a random sample from F_θ, $\theta\in\Theta$ (regular or non-regular). Consider the problem of testing

$$H_0:\ \theta=\theta_0 \qquad \text{vs} \qquad H_1:\ \theta\ne\theta_0.$$

We consider as a test statistic

$$Z_n=\frac{T_n-\theta_0}{s_n}.$$

- T_n is an estimator of θ (MLE).
- s_n is an estimator of $\mathrm{VAR}(T_n)$ such that

$$\frac{\mathrm{VAR}(T_n)}{s_n}\ \xrightarrow{p}\ 1 \qquad \text{as } n\to\infty.$$

- Under H_0, $Z_n\xrightarrow{w}Z\sim\mathcal{N}(0,1)$ as $n\to\infty$.

We reject H_0 iff $Z_n<-z_{\alpha/2}$ or $Z_n>z_{\alpha/2}$, for some fixed α, where z_α is the quantile of order α of the standard normal distribution. Under H_0,

$$\Pr\big(Z_n<-z_{\alpha/2}\ \vee\ Z_n>z_{\alpha/2}\big)\ \to\ \Pr\big(Z<-z_{\alpha/2}\ \vee\ Z>z_{\alpha/2}\big)=\alpha,$$

so the asymptotic type I error probability equals α.

8. **Power of the test.**
Consider an alternative point $\theta\in\Theta$, that is $\theta\ne\theta_0$.

$$Z_n=\frac{T_n-\theta_0}{s_n}=\frac{T_n-\theta_0+\theta-\theta}{s_n}=\frac{T_n-\theta}{s_n}+\frac{\theta-\theta_0}{s_n}.$$

Assume

$$\frac{T_n-\theta}{s_n}\ \xrightarrow{w}\ Z\sim\mathcal{N}(0,1) \qquad \text{as } n\to\infty,$$

and typically $s_n \xrightarrow{p} 0$ as $n \to \infty$ (standard-error scaling). Hence

$$\frac{\theta - \theta_0}{s_n} \to +\infty \quad \text{or} \quad \frac{\theta - \theta_0}{s_n} \to -\infty \qquad \text{(depending on the sign of } \theta - \theta_0),$$

so

$$Z_n \xrightarrow{p} \pm\infty \qquad \text{as } n \to \infty \text{ under } \theta \neq \theta_0.$$

Therefore

$$\Pr\big(Z_n < -z_{\alpha/2} \ \vee \ Z_n > z_{\alpha/2} \ \big| \ H_1 : \theta \neq \theta_0\big) \ \to \ 1 \qquad \text{as } n \to \infty.$$

Asymptotically the power of Z_n is 1.

$$\Rightarrow \text{ Wald test: reject } H_0 \text{ if } Z_n < -z_{\alpha/2} \ \vee \ Z_n > z_{\alpha/2}.$$

9. **Example (Wald test for Bernoulli).**
Let $(x_1, \ldots, x_n)$ be a random sample from $X \sim \text{Ber}(\theta)$.

$$H_0 : \ \theta = \theta_0 \qquad \text{vs} \qquad H_1 : \ \theta > \theta_0 \quad \Rightarrow \quad \text{apply a one-sided Wald test.}$$

As an estimator for θ we take the MLE:

$$\hat{\theta}_n = \frac{1}{n} \sum_{i=1}^{n} x_i.$$

$$\sigma_n^2 = \text{VAR}(\hat{\theta}_n) = \frac{1}{n}\,\theta(1-\theta).$$

By CLT,

$$\frac{\hat{\theta}_n - \theta}{\sigma_n} \ \xrightarrow{w} \ Z \sim \mathcal{N}(0,1), \qquad n \to \infty.$$

Since σ_n depends on θ we cannot use σ_n as scaling, so we use the plug-in estimator

$$S_n = \sqrt{\frac{\hat{\theta}_n(1-\hat{\theta}_n)}{n}}.$$

Because $\hat{\theta}_n \xrightarrow{p} \theta$, by the continuous mapping theorem

$$\hat{\theta}_n(1-\hat{\theta}_n) \ \xrightarrow{p} \ \theta(1-\theta), \qquad \sqrt{\hat{\theta}_n(1-\hat{\theta}_n)} \ \xrightarrow{p} \ \sqrt{\theta(1-\theta)}.$$

Hence

$$\frac{\sigma_n}{S_n} = \frac{\sqrt{\theta(1-\theta)/n}}{\sqrt{\hat{\theta}_n(1-\hat{\theta}_n)/n}} = \frac{\sqrt{\theta(1-\theta)}}{\sqrt{\hat{\theta}_n(1-\hat{\theta}_n)}} \ \xrightarrow{p} \ 1 \qquad \text{as } n \to \infty.$$

We define the test statistic as

$$Z_n = \frac{\hat{\theta}_n - \theta_0}{\sqrt{\hat{\theta}_n(1-\hat{\theta}_n)/n}} \ \xrightarrow{w} \ Z \sim \mathcal{N}(0,1) \quad \text{under } H_0.$$

For level α (one-sided), reject H_0 if

$$Z_n > z_{1-\alpha}.$$